PULASKI · PRE-ALPHA

User Manual

A practical guide to configuring the plant, planning work, running jobs and managing inventory.

This guide explains how to use Pulaski to set up your plant, plan production and follow your orders. It is written for everyone who uses the application. If you are new to manufacturing software, start with Getting started and Key ideas and follow the numbered steps in each section. If you already know MES and ERP systems, skim the key ideas, then jump to the section you need; the In detail parts and the reference tables give the exact rules.

Guide content current through 2026-10-18.

Application screenshots captured from the local demo on 18 October 2026.

Pulaski is pre-alpha. Screens and behaviour may change; consult the in-app messages and current guide for the latest rules.

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1. Getting started

What Pulaski is

Pulaski is a manufacturing execution system (MES) for lines that run continuously or in long batches: extrusion, moulding, compounding and high-speed packaging. It sits between your business system (the ERP, which handles sales, purchasing and accounting) and the machines on the floor.

Today you can use Pulaski to:

- describe your plant: sites, areas, lines and departments;
- describe how products are made (**processes**) and what they are made from (**bills of materials**);
- create **work orders**, which Pulaski turns into **jobs** on your lines;
- plan and sequence those jobs on a live **schedule**, and see when each one is expected to run;
- publish the plan to the floor and freeze it;
- keep **inventory**: receive material lots into stores, write stock up or off, move and hold lots, record what each job used and made, and trace any lot back to what it was made from or forward to what was made from it.

Recording what actually happens on the machines (running jobs, counts, downtime) is still being built; see What Pulaski doesn't do yet.

Opening Pulaski

Open Pulaski in a web browser at the address your administrator gives you (on a developer's machine it is `http://localhost:5090`). Pulaski works best in a current version of Chrome, Edge, Firefox or Safari, on a desktop or a tablet in landscape.

Pulaski asks you to **sign in** with the user name and password your administrator gave you; the first time, you choose your own password. Your **roles** decide which screens you see and what you can change on them. See Users, roles and signing in.

The screen

- **Menu (left).** Two groups of pages:
  - **Operate: Overview**, **Work orders**, **Schedule**, **Floor**, **Inventory**, **Labor**: the pages you use day to day.
  - **Configure: Equipment**, **Materials**, **Processes**, **BOMs**, **Units**, **Reasons**, **Shifts**, **People** (and **Users**, for administrators, with tabs for users, roles and permissions): the setup you do once and change now and then.
  - The ≡ button at the top left collapses the menu to icons, to make room on small screens.
- **You (top right).** Your name and roles; the padlock changes your password, and the door signs you out.
- **Theme (top right).** The sun/moon button switches between light and dark.
- **Main area.** The page you picked.

How saving works

Pulaski saves in two different ways, depending on the screen. Each section says which one it uses.

Screen	How changes are saved
Equipment properties	Straight away, when you press **Enter**, leave the field, or pick a choice. **Esc** undoes the edit before it is saved.
Process editor	Straight away, after every change. The top of the editor says **Saving…** and then **All changes saved**.
BOM editor	When you press **Save draft**. The button says **Saved** when there is nothing left to save.
Forms (new work order, new process, new BOM)	When you press the form's button (**Create**, **Create and edit**).

Screen	How changes are saved
Schedule	Every drag and every button acts straight away. Only cancelling a job asks you to confirm.

Times and time zones

Every time you see or type in Pulaski is in the **site's time zone**, not your computer's. Each site has its own time zone, set when the site is created. Forms remind you which zone they use, for example *Times are in the site's time zone (America/Chicago)*.

When someone else changed the same thing

Several people can work in Pulaski at once. If you save something that another person changed after you opened it, Pulaski refuses your change and tells you, for example *Work order WO-1001 was changed by someone else. Reload and try again.* Nothing is lost: reload the page (or wait for the schedule's next refresh), look at the current version, and make your change again.

Messages and errors

- **Red or yellow boxes** at the top of a page explain why something was refused, in plain words: *Material PRF-5100-BR is not set up to run on EXT-02; add a capability for it first.* Fix what it says and try again.
- **Pop-up notices** at the bottom of the Schedule confirm actions and warn about new problems; they disappear after a few seconds, or close them with ×.

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2. Key ideas

You will meet these words throughout Pulaski. Most follow international standards for manufacturing systems (ISA-95 and ISA-88), so they match what other systems and consultants use.

Your plant: equipment

Everything physical is **equipment**, arranged in a tree from the top of the company down to individual machines:

```
Enterprise          DEMO     the company
└─ Site             PLT1     one plant, with its own time zone
   ├─ Area          EXT      Extrusion area (a department)
   │  ├─ Line       EXT-01   Extrusion line 1
   │  │  ├─ Unit    EXT-01-EXT   the extruder
   │  │  └─ Unit    EXT-01-SAW   the saw
   │  └─ Line       EXT-02   Extrusion line 2
   └─ Area          PKG      Packaging
      └─ Line       PACK1    Packer 1
```

Two settings on equipment matter most for planning:

- **Can run jobs:** work actually runs here. Lines (and cells) have this.
- **Schedulable:** work can be *planned* to it.

A line is both. A **department** (for example the Extrusion area) is schedulable but doesn't run jobs itself: you can plan work to "Extrusion, by Monday", and the floor decides which of its lines runs it. Pulaski calls this a **group**; see Planning to a department.

What you make: materials

A **material** is anything you make or use: a finished profile, a resin, a dry blend, a pallet, an end cap. Each material has a code (its item number), a base unit (metres, kilograms, each) and usually a **material class** that groups similar materials, such as *Extruded profiles*. Materials are kept on the Materials page; those your ERP sends keep their code, name, base unit and type in the ERP.

Capabilities say which material can run on which line, and how fast (its ideal rate). Pulaski never puts a material's job on a line without a capability for it. You add them on the material's page (Where it can run).

Units of measure

Every quantity has a **unit**: kilograms, metres, each, boxes. Each material has a **base unit**, the one its stock is counted in. You don't have to use it everywhere: you can order a profile in feet even though it is stocked in metres. Pulaski converts:

- **Standard units** of the same kind convert for every material: kg ↔ lb, m ↔ ft, ea ↔ dozen.
- **A material's own conversions** cover the rest: 1 m of a profile weighs 1.6 kg; 1 box holds 500 end caps; 1 bag holds 25 kg of resin.

How you make it: processes

A **process** (a "route") is a reusable recipe for *where* an order runs: "Extrude and Pack" runs operation 10, *Extrude*, on EXT-01 and EXT-02, then operation 30, *Pack*, on PACK1. Each operation names one or more lines or departments, each taking a share of the quantity. Processes are versioned: you edit a draft and release it, and orders are made from the released version.

What you make it from: bills of materials

A **bill of materials (BOM)** lists what one material is made from: "1000 m of door frame profile takes 1450 kg of blend and 90 kg of capstock at extrusion, then 84 end caps at packing". When you create an order, Pulaski uses the BOM to work out how much of each component every job will need.

Orders and jobs

- A **work order** is a request to make a quantity of one material, by a due date, using one process.
- A **job** is the piece of a work order that runs on one line (or waits on one department). One order usually has several jobs: one per line in each operation.
- Later, a **run** will record the actual execution of a job.

What you have: lots and locations

A **lot** is one batch of a material with its own number: a delivery of resin, a pallet of profile. Pulaski keeps every lot's quantity, where it is (its **location**: a store, a staging area, a hopper on a line) and its **status** (*available*, *on hold*, *in quarantine*, or used up). Every change is recorded, so each lot has a full history, and Pulaski knows which lots were made from which (**genealogy**). Some materials, such as end caps or stretch wrap, aren't tracked by lot: Pulaski keeps one **balance** of them per location instead. Your ERP still holds the financial stock figures; Pulaski tells it about every movement.

Planning: you decide where and in what order; Pulaski works out when

This is the most important idea in the schedule:

- **You own the line and the sequence**: which line each job is on, and what runs before what.
- **Pulaski owns the times.** It calculates when each job will start and finish from its quantity, the line's rate, the shift calendar and the changeovers between products. This calculation is the **projection**, and it is redone every time the schedule is read, so it always reflects the latest situation.

You never drag a job to a time. You drag it to a place in a line's sequence, and the times follow. Pulaski never moves jobs by itself and never chooses lines for you.

Plan, dispatch, firm

A work order goes through three planning stages:

1. **Planned:** you are still arranging it. Its jobs can move freely.
2. **Dispatched:** you have published it to the floor. Pulaski records each job's projected times as the **baseline**, the plan you will be measured against.
3. **Firmed:** the plan is frozen. Firmed jobs keep their start time.

Then execution: **in progress**, **completed**, **closed**. An order can be **cancelled** at most stages. See Work order statuses.

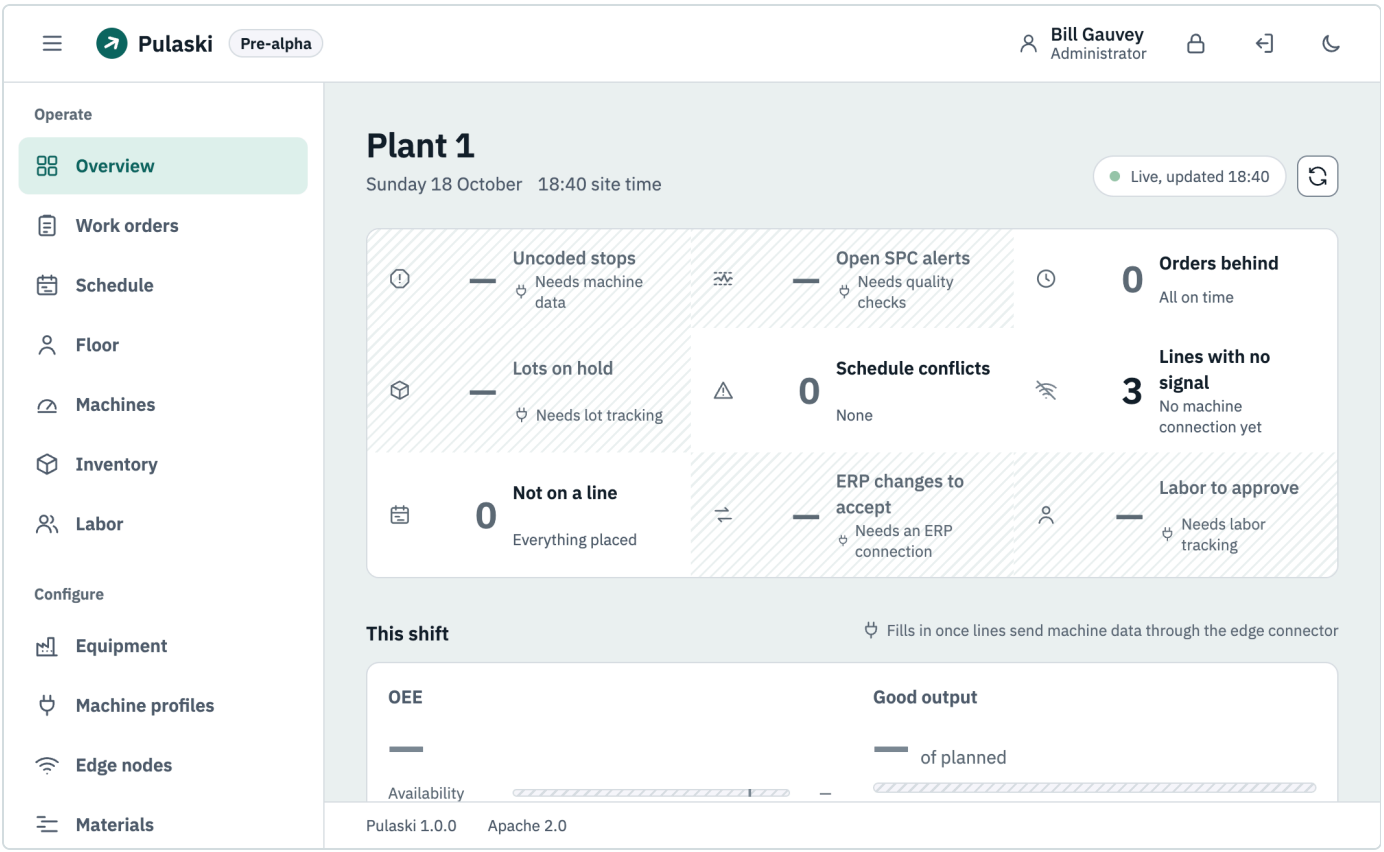
3. Setting up: what to do first

If you are starting from an empty Pulaski, work through this list in order. Each step depends on the ones before it.

#	What	Where	Why
1	The enterprise and at least one site, with its time zone	Equipment	Everything else belongs to a site.
2	Areas (departments) and lines under the site	Equipment	Jobs are planned on lines and departments.
3	Material classes, then materials (one by one, from a CSV file, or from the ERP)	Materials	Orders and BOMs need materials.
3a	Units, and conversions for materials you order or use in other units	Units	So orders, BOMs and rates can use any unit that converts.
4	Capabilities: which material runs on which line, and how fast	The material's page	Without one, a line can't take a material's job.
5	Shift schedules, and which one each site, area or line follows (optional)	Shifts, then Equipment	Without one, Pulaski assumes lines run around the clock and marks them **24/7**.
5a	Crews and people, if you record labor (optional)	People	So supervisors can check crews onto lines, and shifts can name their crew.
6	Processes: the routes, what they make, released	Processes	Every work order is made from a process.
7	BOMs for the materials you make (optional, recommended)	BOMs	Without a BOM, an order plans no components.
8	Work orders	Schedule → New work order	Now you can plan.
9	Inventory locations: stores, staging areas and feed points, with **Holds inventory** ticked	Equipment	Lots have to be somewhere.
10	Reason codes for write-ups, write-offs and holds, and utilization reasons for machine time	Reasons (a starter set comes with `db/seed_reasons.sql`)	Every adjustment and hold needs a reason; machine profiles map stop codes to utilization reasons.
11	Machine profiles, and which one each machine uses	Machine profiles, then Equipment	So the machine's state values and stop codes mean something in Pulaski.
12	Edge nodes, their connections, and tags binding machine signals to equipment (optional)	Edge nodes	So Pulaski hears machine state and counts.

The demo data that comes with Pulaski already contains steps 1 to 4, and 9 (plant PLT1 with EXT-01, EXT-02 and PACK1, the stores RM-STORE and staging FG-STAGE, 200 materials, and the standard units with conversions for the demo materials), so you can start at step 6.

The Overview page



Overview: plant status, attention items and line strip. Local demo screen; details vary by site, data and role.

Operate → Overview is the home page: a live board for the plant. In a few seconds it answers *which lines are running, which aren't and why, and what needs a person*. It refreshes itself every 30 seconds; the pill at the top right says when it was last updated (and turns grey, with a warning, if Pulaski can't reach the server for over a minute). The refresh button updates it at once. With several sites, pick one at the top.

Diagonal hatching means "no data yet". Several parts of the board need machine data, quality checks, lot tracking, labor tracking or an ERP connection, none of which Pulaski collects yet. Those parts already sit where they'll be, hatched and greyed, with a plug symbol saying what will fill them, so the page won't move around when the data arrives.

Needs attention (the grid at the top) counts what needs a person, most urgent first. A count above zero is marked with a coloured bar and icon; click it to go to the list behind it.

Cell	Counts	Today
Uncoded stops	Stops this shift with no reason given	Needs machine data
Open SPC alerts	Control-chart alerts to respond to	Needs quality checks
Orders behind	Open orders projected late or slipping past their baseline	Live → Work orders
Lots on hold	Lots on hold or in quarantine	Needs lot tracking
Schedule conflicts	Open orders with a schedule warning or an overlap	Live → Schedule
Lines with no signal	Lines that should report their state but don't	Live: every line, until machines are connected
Not on a line	Planned orders with quantity not yet placed	Live → Schedule
ERP changes to accept	Orders the ERP changed after dispatch	Needs an ERP connection
Labor to approve	Labor sessions from the last shift to approve	Needs labor tracking

This shift will show OEE (with availability, performance and quality against target), good output against plan, scrap and downtime with its biggest reason. All four need machine data.

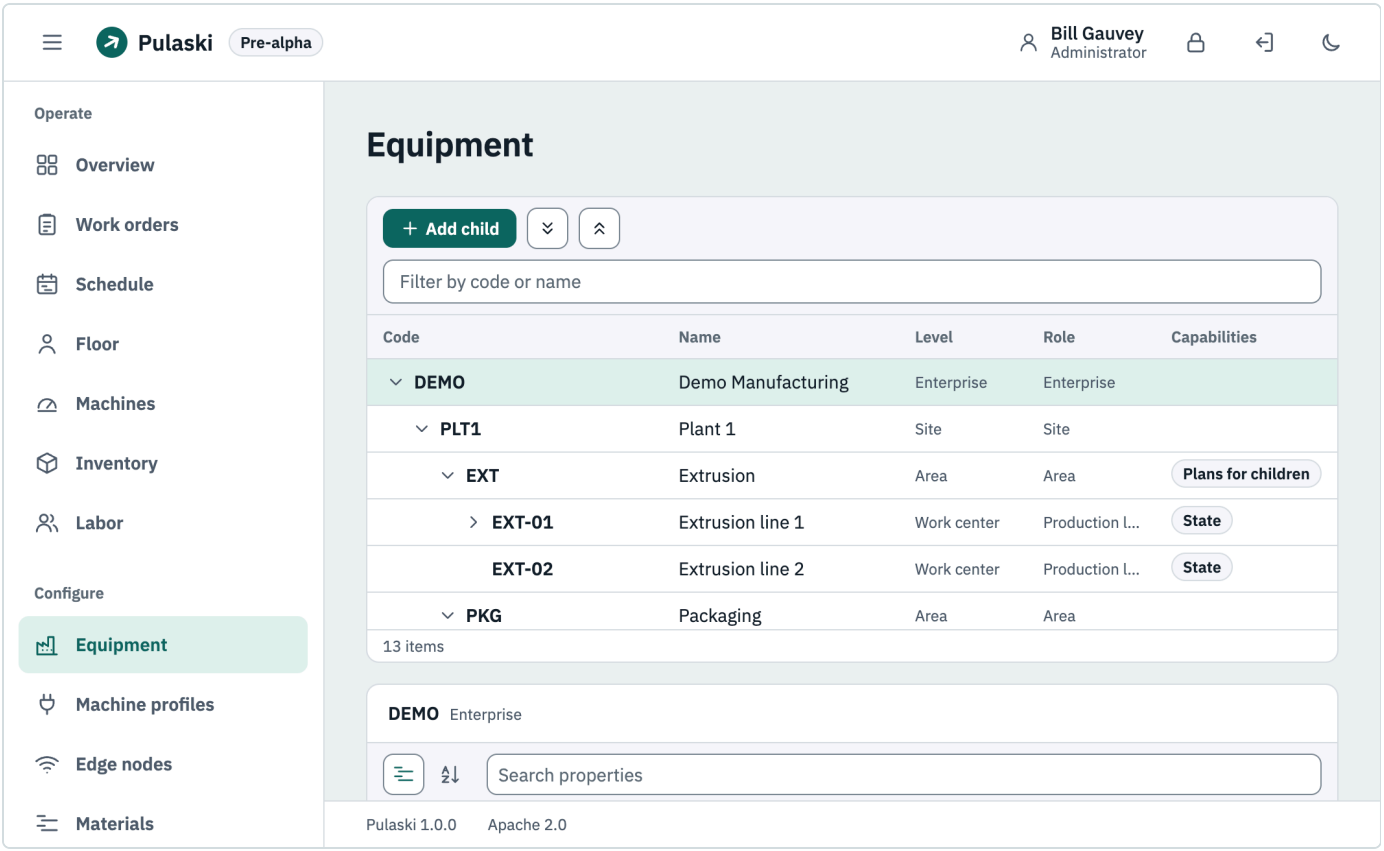
Lines has one row per line:

- **State:** the machine state. Until the line is connected it says **No signal**.
- **Now or next:** the job the schedule has on the line now, or the next one and when it starts, with how much of it is made.
- **The strip** covers six hours back to eighteen hours ahead, with a black line at *now*. The top lane will show what the line actually did (hatched while there's no signal); the bottom lane shows the planned jobs: outlined for planned, solid for dispatched or firmed, hatched for changeovers. Hover over a job to see it; click it to open its order on the Schedule. If nothing is planned in the window, the lane says when the next work starts.
- **OEE** for the line, once there's machine data.
- **Flags** at the end: a red mark if work overlaps on the line, an amber clock if a job has a schedule warning (hover for the messages).

At the bottom, **Work orders** counts orders at each stage, Planned → Dispatched → Firmed → In progress → Completed (click one to list them), and **Starting next** lists the next six jobs to start across the plant.

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4. Equipment



Equipment: the plant hierarchy and equipment properties. Local demo screen; details vary by site, data and role.

Configure → Equipment. This is where you describe your plant as a tree.

The screen

- **Left: the tree.** Each row shows the code, name, level, role and chips for its settings: **State** (tracks machine state), **Schedulable** or **Plans for children** (a department), **Runs jobs**, **Inventory**, **Mixes lots** (lots are indistinguishable), **One material**, and **Inactive** in grey.
  - Click the arrow beside a row to open or close it, or use **Expand all** / **Collapse all**.
  - Type in **Filter by code or name** to find equipment; matching rows stay visible with their parents.
  - Arrow keys move through the tree when it has focus.
  - The bottom line counts the items (and the matches while filtering).
- **Right: the properties** of the selected equipment, grouped into *Identity*, *Classification*, *Capabilities*, *Site*, *Status*, *Machine data* (equipment that tracks state), *Working time* and *Hierarchy*.
  - Click a property to see what it means in the description area at the bottom.
  - Switch between grouped (**Categorized**) and **Alphabetical** order, or search the properties by name.

Add equipment

1. In the tree, click the equipment the new one goes **under** (the parent). For the very first item there is nothing to select: the button reads **Add enterprise**.
2. Click **Add child**. A new row marked **Unsaved** appears under the parent, and its properties open on the right.
3. Fill in the properties:
   - **Code** (required): a short, unique business key such as `EXT-03`. It can't be changed later.

- **Name** (required): what people call it, such as *Extrusion line 3*.
- **Level**: where it sits in the tree. Only levels below the parent are offered.
- **Role**: what kind of equipment it is at that level, for example *Production line* or *Storage zone*. Picking a role sets sensible capability defaults (below), which you can change.
- **Time zone** (sites only, required): for example `America/Chicago` or `Europe/Berlin`.
- The **capabilities** (below).

4. Click **Create** at the bottom of the properties, or **Cancel** to throw the new row away.

Change equipment

Select it and edit a property on the right. Text saves when you press **Enter** or leave the field; a choice saves as soon as you pick it. **Esc** undoes a text edit before it saves. **Code**, **Level**, **Role** and the site's **Time zone** can't be changed after creation.

Capabilities

Setting	Turn it on when…	Default for
Tracks state	Machine signals for this equipment should be recorded (running, stopped…), for downtime and OEE.	Lines, cells, units, machines
Schedulable	Work should be planned to it.	Lines, process cells, production units
Can run jobs	Work actually runs on it.	Same as Schedulable
Holds inventory	Material lots can be stored or loaded here.	Feed points, storage zones and units
One material at a time	Shown once **Holds inventory** is ticked. Another material can arrive only when the quantity of the one there is 0 (stock on hold still counts). It can't be turned on while two materials are there.	Silos and tanks dedicated to one material
Lots are indistinguishable	Shown once **Holds inventory** is ticked. Lots here mix and can't be told apart, like a silo or tank: everything that arrives joins one balance per material. See Inventory. It can't be turned on while separate lots are there; merge or move them first.	Silos, tanks, mixing hoppers

A location holding stock can't be deactivated or stop holding inventory: move, use or write off the stock first.

Lines vs departments. A line has *Schedulable* and *Can run jobs*. A department (an area that you plan to, with lines below it) has *Schedulable* only; the tree shows it as **Plans for children**. A line can also have *Can run jobs* without *Schedulable*: it then only takes jobs from a department above it.

Machine data

Equipment with **Tracks state** set shows a **Machine data** group:

1. **Machine profile**: how its controller reports state and stop codes. Choose one of its site's machine profiles, or *None*.
2. **Micro-stop threshold**: stops shorter than this many minutes are micro-stops. They are logged but not queued for a reason, and they don't count against availability. Default 5 minutes; 0 to 60, in steps of half a minute. An extruder may use 2, a pack cell more.
3. **State from**: where its state comes from.
   - *Its own state tag* (default).
   - *Worst of its machines (series)*: for a line whose machines run one after another, so one stopped machine stops the line.
   - *Best of its machines (parallel)*: for machines running side by side, so the line runs while any of them does.

    - *Its constraint machine*: the line's state is one machine's. A **Constraint machine** row appears; Pulaski first picks the first machine below that tracks state, and you can choose another.

Each saves as soon as you change it. The profile reads the machine's data as it arrives through an edge node; the threshold decides which stops are micro-stops, and *State from* how a line shows on Machines.

Deactivate equipment

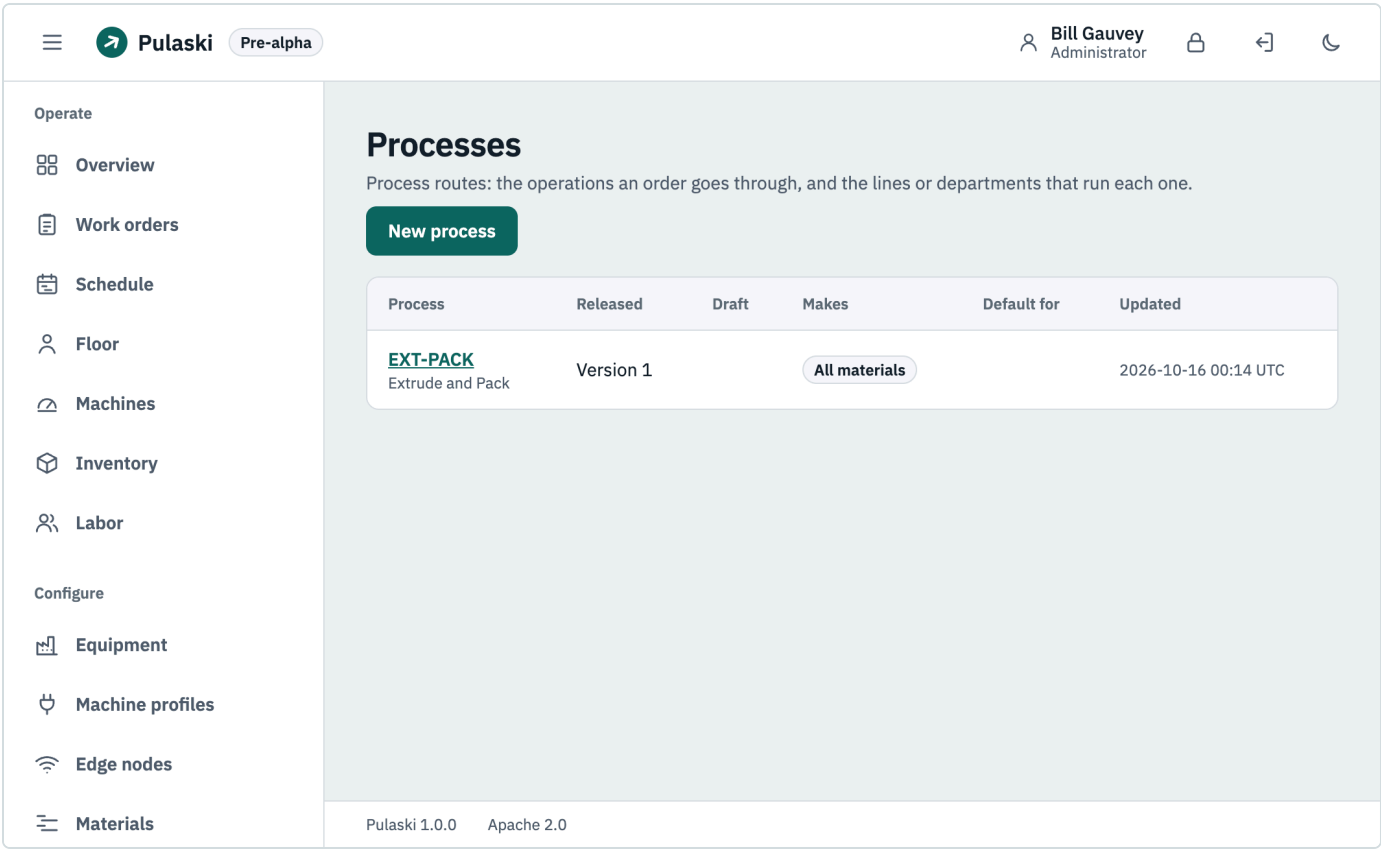
Set **Active** to *false*. Inactive equipment stays in the tree and in history, greyed out, but can no longer be used for new work. Pulaski never deletes equipment.

In detail

- The levels, top to bottom: enterprise, site, area, work center, work unit, equipment module, control module. A child always sits below its parent's level.
- Roles allowed at each level: work center: production line, process cell, production unit, storage zone; work unit: work cell, unit, machine, storage unit; equipment module: machine, feed point, module; control module: module.
- Codes are unique within a site.
- The **Working time** group: **Shifts** chooses when it works for planning (*Inherit from parent*, one of its site's shift schedules, or *Around the clock*), and **Shifts in effect** shows what it ends up with and where from, such as *2X8 · Two shifts, from PLT1*. See Shift schedules. Changing **Shifts** needs permission to maintain shift schedules, not equipment.
- **Machine data** needs permission to maintain equipment. A machine profile must be active and of the equipment's own site, and only equipment that tracks state takes one. A constraint machine must be below the line and track state.
- The **Hierarchy** group shows the parent, the owning site, the number of children and Pulaski's internal id (useful for support and the API).

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5. Processes



Processes: released routes and their operations. Local demo screen; details vary by site, data and role.

Configure → Processes. A process describes where an order runs: its operations, the lines or departments that run each one, and how the operations follow each other.

Example

"Extrude and Pack" (EXT-PACK):

- **Operation 10, Extrude**, runs on EXT-01 (50%) and EXT-02 (50%).
- **Operation 30, Pack**, runs on PACK1 (100%), and starts once extrusion has started.

A 2,400 m order made from it gets three jobs: 1,200 m on EXT-01, 1,200 m on EXT-02 and 2,400 m on PACK1.

The process list

Each row shows the process code and name, its **Released** and **Draft** versions, what it **Makes**, and the materials it is the **Default for**. Click the code to open the editor.

Create a process

1. Click **New process**.
2. Pick the **Site** (only shown when there are several), and enter a **Code** (for example `EXT-PACK`) and a **Name** (*Extrude and Pack*).
3. Click **Create and edit**. The editor opens on an empty draft, version 1.

The editor

- **Top:** the versions (click one to view it), the save state, and the release button.
- **Left: Lines and departments** of the site. Hide or show it with the panel button at the top left of the canvas.
- **Middle: the canvas**, where operations are boxes and links are arrows.
- **Right: properties** of what is selected: an operation, a link, or (with nothing selected) the process itself. Hide or show it with the panel button at the top right.
- **Bottom: a table** of every operation, the equipment it runs on, each share, the total, and what it follows.

A draft saves after every change; you never need a save button.

Build the route

1. **Add an operation.** Either drag a line or department from the left onto an empty part of the canvas (a new operation is created with it), or click **Add operation**.
2. **Name and number it.** With the operation selected, set its **Number** (10, 20, 30… leave gaps so you can insert steps later), **Name** and, optionally, **Description**.
3. **Add more equipment to it.** Drag another line or department onto the operation box, or use **Add a line or department…** below its properties. Each one appears with its **share**.
4. **Set the shares.** Edit each share in the properties, or click **Split evenly**. The shares of an operation must total exactly **100%**. Each entry becomes its own job for its share of the quantity. Remove an entry with its × in the properties.
5. **Link the operations.** Drag from the small handle on the right edge of an operation to the operation that follows it. An arrow appears with a label such as *after start*.
6. **Set how the link works.** Click the arrow's label and set:
   - **Starts:** *after start* (material flows straight through, so both run together, as with extrude and pack) or *after end* (the next operation waits for the earlier one to finish: a hand-off).
   - **Lag:** extra minutes to wait after that start or end.
7. Move boxes by dragging their header, or click **Arrange** to lay them out automatically.

Delete an operation or link by selecting it and clicking **Delete operation** / **Delete link**.

Say what the process makes

With nothing selected (click an empty part of the canvas), the properties show the process itself:

- **All materials:** *true* if it can make anything. Otherwise, list what it makes with **Make a material class…** (every material in that class, for example *Extruded profiles*) and **Make a single material…**. Remove entries with ×.
- **Default for a material…:** makes this the process that new orders for that material pick automatically. A material can have one default.
- **Active:** an inactive process can't be picked for new orders.

A process that makes nothing yet is flagged; it won't be offered when creating orders.

Release it

While the draft has problems, a yellow box lists them (click one to select the operation it is about), and the release button is disabled. Typical problems: an operation with no equipment, shares that don't add up to 100%, operations not linked to the rest, a loop in the links.

1. Click **Release version 1…**.
2. Optionally type what changed, then click **Release**. (**Not yet** goes back.)

The released version is what new orders are made from. It can never be changed.

Change a released process

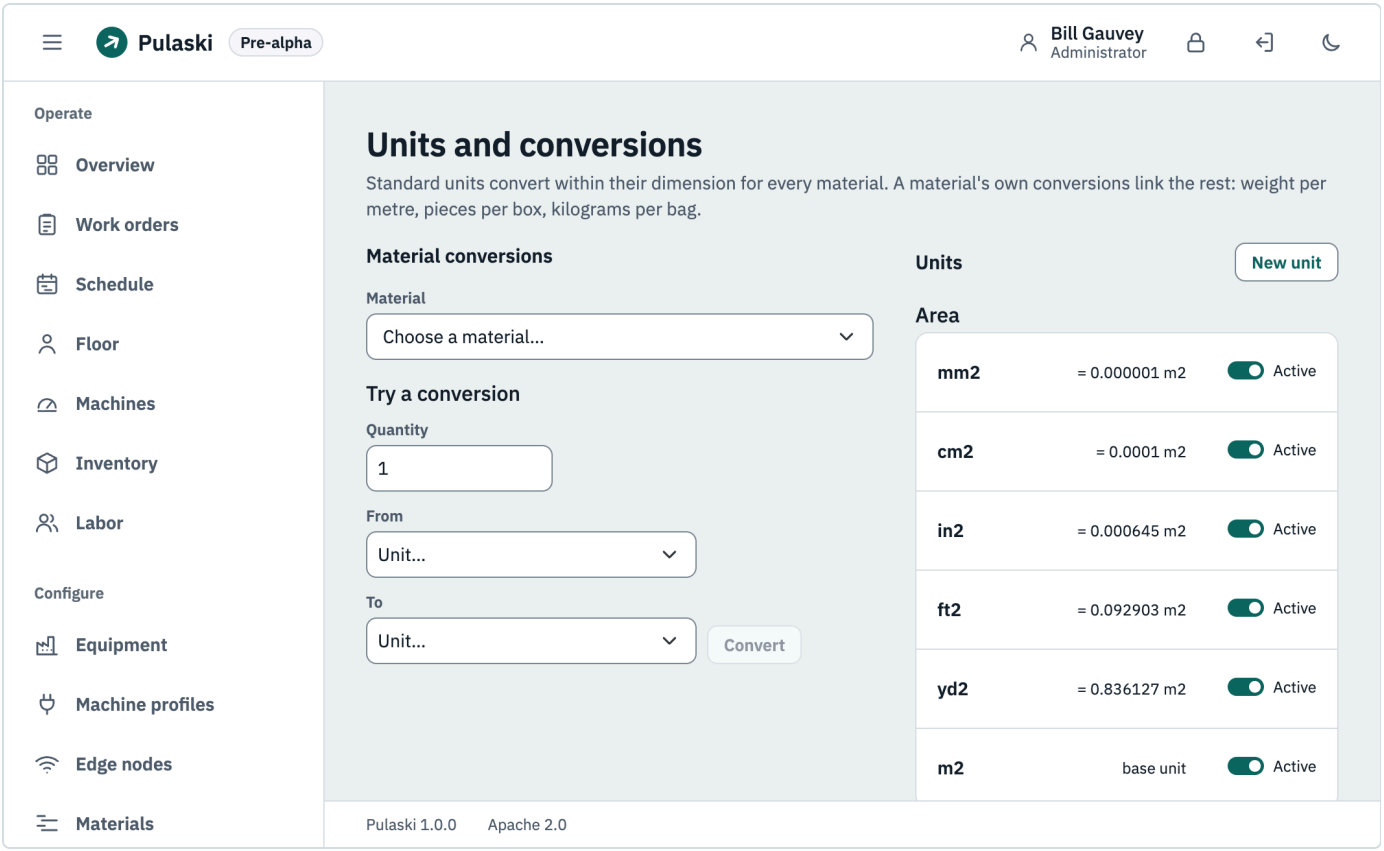
1. Open the process. If there is no draft, click **Change: start version 2** (the number follows on). This copies the released version into a new draft.
2. Edit the draft and release it. The previous version becomes **obsolete**.

Orders already made keep the version they were made from. If a draft already exists, click **Open the draft** instead.

In detail

- Operations with no path of links between them run in parallel; links make them sequential. Links must flow one way; a loop is refused.
- A released process with more than one operation must be connected.
- Only lines and departments of the process's own site can be used; machines and other equipment that neither runs nor plans jobs are refused.
- Today the link's *after start/after end* and lag set the **earliest** time each operation's jobs may start, counted from the order's start. The schedule doesn't yet wait for the earlier operation to actually start or finish. See What Pulaski doesn't do yet.

6. Units and conversions



Units: unit definitions and material conversions. Local demo screen; details vary by site, data and role.

Configure → Units. Here you see every unit Pulaski knows, and say how a material converts between units that aren't of the same kind.

Why it matters

People work in different units. A customer orders 10,000 ft of profile, the BOM is written per 1000 m, the line's rate is in metres per hour, and the blend arrives in 25 kg bags. Pulaski converts between all of these, as long as it knows how. When it doesn't, you'll see a message such as *ft doesn't convert to kg for this material; add a conversion for it*, and this page is where you fix it.

Two kinds of conversion

1. **Standard conversions** come with the unit. Units of the same kind always convert, for every material, by fixed factors: 1 lb = 0.45359237 kg, 1 ft = 0.3048 m, 1 dozen = 12 each. You never enter these.
2. **Material conversions** are true for one material only:
   - between kinds: 1 m of PRF-5100-BR weighs **1.6 kg** (a lighter profile weighs less);
   - to a packaging unit: 1 **box** of end caps holds **500 each**; 1 **bag** of resin holds **25 kg**.

A material conversion works both ways, and combines with the standard ones. With *1 m = 1.6 kg*, Pulaski also knows that 1 ft = 0.48768 kg and 1 ft = 1.07515 lb.

The units list (right side)

Units are grouped by kind: **Area**, **Count**, **Length**, **Mass**, **Packaging (per material)**, **Time** and **Volume**; within each kind, smallest first. For each unit you see its code, name and what one of it is in the kind's base unit (= *0.3048 m*). The base unit itself says **base unit**; packaging units say **per material**.

- **Rename** a unit by typing over its name; it saves when you leave the field.
- **Switch a unit off** with its **Active** switch. Inactive units disappear from the lists where you choose a unit, but quantities already in them still convert.
- **Add a unit** with **New unit**: enter the **Code** (for example `bdl`), the **Name**, the **Kind** and, for anything but packaging, how many of the base unit one of it is. Click **Create**.

A unit's code, kind and factor can't be changed after it is created, because quantities are stored in it. If one is wrong, switch it off and create a new one.

Add a conversion for a material

1. In **Material conversions** (left side), choose the **Material**. The note under it says which unit it is stocked in.
2. In the row under the table, fill in **1 [unit] = [quantity] [unit]**. For example: 1 `m` = `1.6` `kg`, or 1 `box` = `500` `ea`.
3. Click **Save conversion**. It appears in the table with the reverse worked out (*1 kg = 0.625 m*).

To change a conversion, save it again with the new quantity (in either direction; Pulaski replaces the old one). To remove one, click **Remove**.

Pulaski refuses a conversion when:

Message says…	Because…
…already convert by their standard factors	Both units are of the same kind (kg and lb): they already convert for every material.
…already converts to … through its other conversions; change one of those instead	The material can already get from one unit to the other. Two routes could give two different answers, so Pulaski allows only one. Change the existing conversion instead.

Try a conversion

Under the conversions, **Try a conversion** works out any quantity: enter a **Quantity**, choose **From** and **To**, and click **Convert**. With a material chosen, its own conversions are used too, so you can check, for example, what one bundle of profile weighs in pounds. Without a material, only standard conversions work.

Where conversions are used

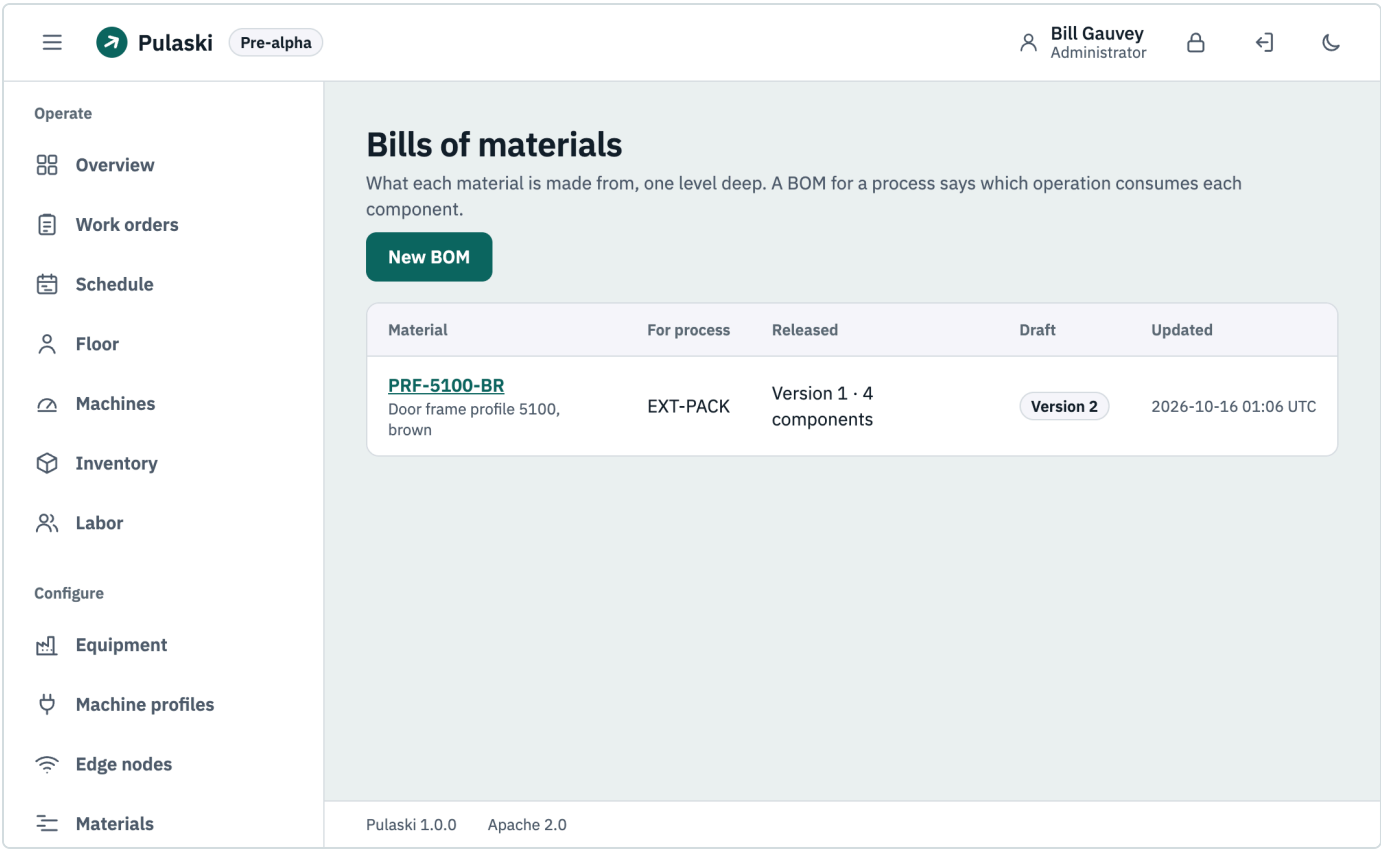
Where	What converts
Creating a work order	The order may be in any unit that converts to the material's base unit. It is converted to the BOM's unit to plan the components.
BOMs	The BOM's unit must convert to the material's base unit, and each component's unit to that component's base unit, before the BOM can be released.
The Schedule	A job's length is worked out from the line's rate even when the job is in another unit (an order in feet on a line rated in metres per hour). The side panel shows the rate in the job's unit.

In detail

- Base units: kg (mass), m (length), m² (area), L (volume), ea (count), s (time). The standard factors are the exact legal definitions, not approximations.
- Conversions are kept exact along the way; a converted quantity is rounded to six decimal places.
- A material's conversions can never offer two routes between two units, so every conversion has exactly one answer.

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7. Bills of materials (BOMs)



BOMs: component planning for a released process. Local demo screen; details vary by site, data and role.

Configure → BOMs. A BOM lists what one material is made from, so that every order plans the components each job will use.

Ideas to know first

- **One level.** A BOM lists only the direct components. If your profile is made from a dry blend, and the blend from resin and additives, the profile's BOM lists the *blend*, and the blend has its own BOM. Each is made on its own orders.
- **For a process, or for any process.** A BOM can be tied to one process. Then each component can name the **operation** that consumes it: blend at *Extrude*, end caps at *Pack*. A BOM for "any process" is used when the order's process has no BOM of its own; all its components go to the first operation.
- **Quantities are per a base quantity.** For example, "per 1000 m of PRF-5100-BR". Orders can be in any unit that converts to it.
- **Versions**, as with processes: edit a draft, release it, start a new draft to change it.

Create a BOM

1. Click **New BOM**.
2. Pick the **Material** the BOM is for (raw materials and consumables aren't offered, since you don't make them).
3. **For process:** leave **Any process**, or pick a process. Only processes that make this material are offered.
4. Click **Create and edit**.

There can be one BOM per material and process, plus one for any process.

Fill it in

1. **Quantities are per:** enter the base quantity, for example `1000`, and choose its unit (it starts as the material's base unit). It must convert to the material's base unit. The link beside it opens the material's unit conversions.
2. Click **Add component** for each line, and fill in:

Field	What to enter
Line	The line number. New lines go up in tens (10, 20, 30) so you can insert between them.
Component	The material consumed. Its unit is filled in for you.
Quantity	How much is used per the base quantity.
Unit	The unit of that quantity.
Scrap %	Extra to plan for losses. `2` plans 2% more.
Consumed by	How it will be taken from stock once execution is built: **Backflush on count** (automatically, in proportion to good output), **Time window (loaded lot)** (whatever lot is loaded at the feed point while the job runs, typical for resin and blend), **Declared by operator** or **Scanned lot**.
Operation	Only on a BOM for a process: which operation consumes it. **First operation** if you leave it.

3. Remove a line with its red ×.
4. Click **Save draft**. A yellow box lists anything that stops the release, and the lines concerned are highlighted.

Release it

When the draft is saved and has no problems, type an optional **Release note** and click **Release version N**. From then on, new orders for that material (with that process) plan their components from it.

Change it

Open the BOM and click **New draft**. It copies the released version; edit, save, release. The old version becomes obsolete. Orders already made keep the components they were planned with.

Worked example

BOM for PRF-5100-BR, for EXT-PACK, **per 1000 m**:

Line	Component	Quantity	Scrap	Consumed by	Operation
10	CMP-PVC-PRF-WH (white blend)	1450 kg	2%	Time window	10 Extrude
20	CMP-PVC-PRF-CAP (capstock)	90 kg	0	Time window	10 Extrude
30	PKG-ENDCAP	84 ea	0	Backflush	30 Pack
40	PKG-PAL-LONG	0.5 ea	0	Declared	30 Pack

A 2,400 m order, extruded 50/50 on EXT-01 and EXT-02, plans:

- Blend: 2400 ÷ 1000 × 1450 × 1.02 = **3,549.6 kg**, so 1,774.8 kg for each extruder job.
- Capstock: 2400 ÷ 1000 × 90 = **216 kg**, so 108 kg each.
- End caps: **201.6**, all on the PACK1 job. Pallets: **1.2**, on the PACK1 job.

In detail

- A material can't be its own component. Components must exist and be active to release.
- To release, each component's unit must convert to that component's base unit (so its use can be taken from stock later), and the BOM's unit must convert to the material's base unit. The yellow box says which conversion is missing.
- Operation numbers must exist in the process's **released** version. A draft may name an operation that isn't released yet, but can't be released until it is.
- Each component's total is rounded to six decimal places. The jobs of an operation share it in proportion to their quantities, and the last job takes any rounding remainder, so the jobs always add up to the total.
- Planned components never change after the order is made, even if the BOM changes later.
- A job added to the order later (from the Schedule, or the replacement when a job is cancelled and replaced) gets its components too: those of its operation, for its own quantity, from the BOM version the order was planned with.

When the process changes

A released BOM names operations by number. If the process is later released with different operations (Pack renumbered from 30 to 40, say), the BOM still names 30. Pulaski doesn't stop the process release, but:

- the process editor shows which BOMs the release broke;
- the BOMs list and the BOM editor say which operation the BOM names that the process no longer has;
- the new-order form greys out the process for that material, and creating the order is refused with the same reason.

To fix it, open the BOM, start a new draft, move those lines to operations the process has, and release it.

8. Creating a work order

Schedule → New work order (the button at the top right of the Schedule).

You choose *what*, *how much* and *when*. The process you choose decides *where*: each of its operations becomes jobs on its lines or departments.

Step by step

1. **Site:** the plant (the first one is picked for you).
2. **Order number:** your reference, for example `WO-1001`. Unique within the site.
3. **Material produced:** what to make. The unit fills in with the material's base unit.
4. **Process:** the processes at this site that make this material and have a released version.
   - The material's **default** process (marked · *default*) is picked automatically, or the only one if there is just one.
   - A process whose lines can't run this material is **greyed out** with *can't run this material*, and the reason is listed underneath, for example *Operation 10 (Extrude): EXT-02 is not set up to run FLM-HDPE-20-BLK; add a capability for it first.*
5. **Planned quantity** and **Unit:** how much, in any unit that converts to the material's base unit (the hint under the field links to its conversions). For a material made in batches it must be a whole number of batches; the hint shows the batch size.
6. **Start:** when the order may start. No job will be planned before it. It defaults to the next whole hour at the site.
7. **Due:** when it must be finished (optional). Jobs projected to finish later are flagged **late**. If the process uses a department, the due date is **required** (the label says so): it is the department's finish-by time.
8. **Priority:** a number; higher is more urgent. Shown as an arrow on the Work orders screen.
9. **Notes:** anything the floor or other planners should know (up to 4000 characters).
10. Check the preview at the bottom:
    - **Jobs this order will create:** each operation, its line or department, its share and its quantity.
    - **Components:** which BOM will plan the components, or a warning that the material has no released BOM (the order then plans none).
11. Click **Create**. You go to the Schedule with the new order open in the side panel.

What happens

- The order is created as **planned**.
- Each operation's lines and departments get one **job** each, numbered from the order: `WO-1001-30` for an operation with one line, `WO-1001-10-1` and `WO-1001-10-2` for an operation with two.
- Each job goes **at the end** of its line's sequence. Move it on the schedule if it should run sooner.
- Jobs are held until the start time (later for an operation that follows another with a lag).
- A job on a department waits there in a window from its start to the due date, until the floor assigns it to a line.
- If the material has a released BOM, each job's components are planned.

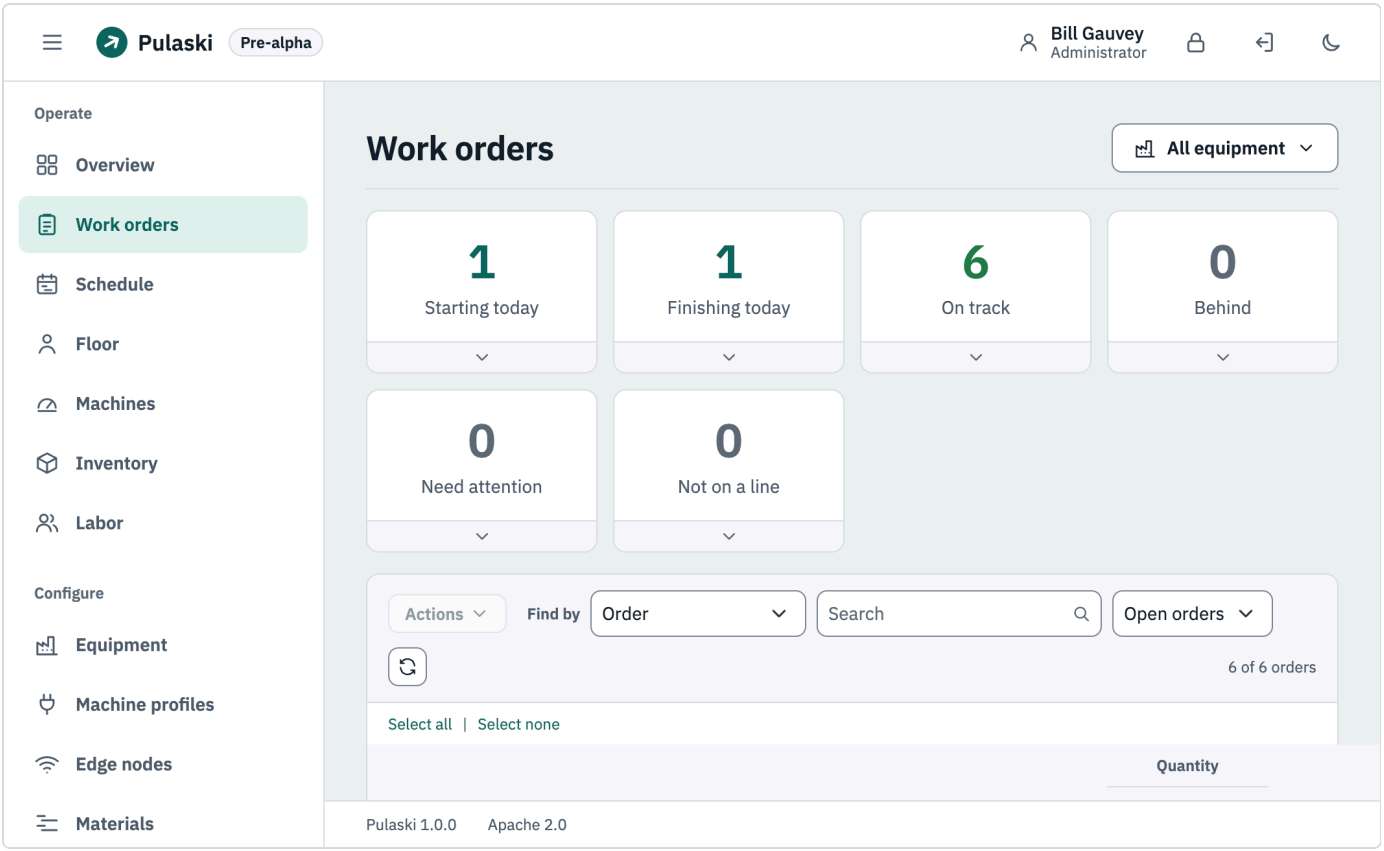
If it is refused

Everything is checked before anything is created, and all problems are listed together. Nothing is created until they are fixed. Common reasons:

Message says…	What to do
…is not set up to run…; add a capability for it first	Ask your administrator to add a capability for the material on that line, or pick another process.
…is a department, so its job needs a finish-by; give the order a due date	Enter a **Due** date.

Message says…	What to do
…doesn't make…	The process's scope doesn't include the material: add it on the process, or pick another process.
…doesn't convert to…, the base unit of…	Enter the quantity in another unit, or add a conversion for the material.
…was changed by someone else / *…was released again*	Try again: the process changed while you were filling in the form.
Duplicate order number	Use another order number.

9. The Work orders screen



Work orders: the list of orders and their current stages. Local demo screen; details vary by site, data and role.

Operate → Work orders. A list of orders with how each one stands: when it is projected to run, how much is made, and whether it is behind.

Choose what you see

- **Equipment picker (top right):** show orders for the whole plant, one site, one department or one line. Search for equipment by typing.
- **Tiles:** counts of orders that are **Starting today**, **Finishing today**, **On track**, **Behind**, **Need attention** and **Not on a line**. Click a tile to show only those orders; click it again (or **Show all**) to clear. Hover over a tile for its exact meaning (see Tile meanings).
- **Find by:** search by **Order**, **Material** or **Line or department**, then type in **Search**.
- **Status: Open orders** (the default), one status, or **All statuses**.
- **Refresh** reloads the list.

Read a row

Column	Shows
Flag	A red mark when the order is behind; hover for why (*Projected to finish 3h after its due date*).
Order	The order number; an up arrow marks priority.
Status	Planned, Dispatched, Firmed…
Material	Code and name.

Column	Shows
ⓘ	Opens the details (below).
Schedule	Projected start and finish.
Quantity	Planned and made so far.
Actions	The row's menu (below).

Details

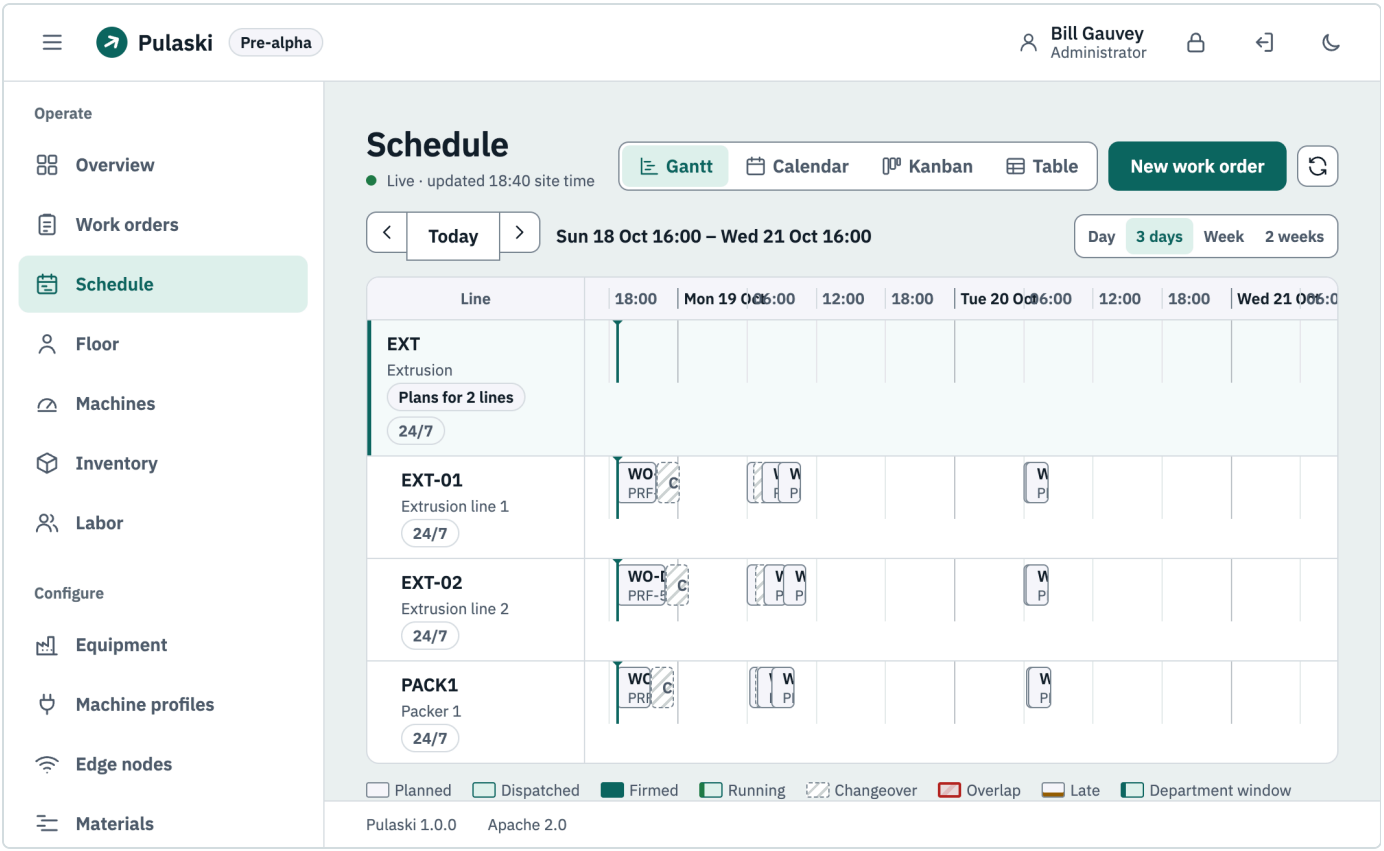
Click ⓘ to open a row. You see the due date, the published baseline, how much is on lines, the process and version, the BOM version, the origin, the notes, every open job with its line, status, projected times and remaining quantity, any warnings, and the **planned components** for each job.

Act on orders

- **One order:** open the row's **Actions** menu (the ⌄ button) and choose **Dispatch**, **Un-dispatch**, **Firm**, **Cancel** and so on (only what the order can do next is shown), or **Show on the schedule**.
- **Several orders:** tick **Select** on each (or **Select all**), open **Actions** at the top of the list and pick an action. Each choice shows how many of the selected orders it applies to (*3 of 5*); the others are skipped. **Cancel…** asks you to confirm.

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10. The Schedule



Schedule: Gantt plan by line, with jobs and changeovers. Local demo screen; details vary by site, data and role.

Operate → Schedule. The live plan: every line and department, every open job and order, and when each is projected to run.

Before you start

- The schedule only shows equipment marked **Schedulable**. If it says *No schedulable lines*, mark your lines as schedulable on Equipment.
- **Live updates.** The board re-reads everything every 10 seconds; the dot and *Live · updated 14:05 site time* at the top show it. Cards that changed since the last update are briefly highlighted, so you notice other planners' changes. Updates pause while you drag. Click the refresh button to update now.
- **Site:** choose the site at the top if you have more than one.
- **Views: Gantt**, **Calendar**, **Kanban** and **Table** show the same plan in four ways. Each view lets you change only what the planner owns (line, sequence, pins, order status).
- **The side panel** opens when you click any job or order, on any view. Everything you can do by dragging, you can also do there, with the keyboard. **Esc** closes it.

Gantt view

Lines down the left, time across.

- **Move through time: Earlier**, **Today** and **Later**; zoom to **Day**, **3 days**, **Week** or **2 weeks**. The vertical line marks *now*.
- **Bars** are jobs, coloured by status (see the legend under the chart). Changeovers between products appear as their own bars, in the changeover style, before the job that needs them. Late and overlapping work is marked too.
- **Line labels** show the code, name and chips: the shift schedule the line follows, such as **2X8** (hover to see where it comes from), **24/7** (no shift schedule, or set to run around the clock) and **Plans for N lines** (a department).

Resequence or move a job: drag its bar. A marker shows the slot it will go into: before the job under the pointer, or at the end of the line. Drop it on another line to move it there. Times then recalculate.

Place an order that isn't on a line yet: planned orders with quantity still to place are in the **To place** tray on the left. Drag one onto a line. While you drag, each line shows a **drop lane** previewing where the new job would land and which jobs it would push later; a line that can't take it shows why. Dropping creates a job for the unplaced quantity.

Departments: a department's row shows each waiting job as a band over its window, with the fastest line's estimate. Each line that could run it shows a faint **ghost** ("~14h here"). Drag the band or a ghost onto a line to assign the job there. Dragging an order from the tray onto a department opens the planning form with a suggested window.

Calendar view

A month of work by day; a job that spans several days appears on each. The number in each day counts the jobs running that day.

- **Hold a job until a later day:** drag a planned job to that day. It won't start before then.
- Dragging to an earlier day only clears a hold: to run something sooner, change its place in the line on the Gantt.

Kanban view

Work orders as cards in columns by status: Planned, Dispatched, Firmed, In progress and Completed. Turn on **Closed and cancelled** to show those columns too. Click a card to open the order.

- **Change an order's status:** drag its card to another column: **Dispatch**, **Un-dispatch**, **Firm** and so on. Columns it can't move to dim while you drag.

Table view

Every open job in one sortable list.

- Type in **Find job, order or material** to filter; click a column heading to sort.
- Edit in place: **Lock**, **Hold until** and **Finish by**.
- Warnings and overlaps show as chips; hover for the message.

The side panel: a job

- **Alerts** at the top: overlaps and warnings for this job.
- **Facts:** projected (or window) times and duration, the department it was planned to, the finish-by, the baseline, its work order, material, quantity still to make, the rate used and where that rate came from, the due date.
- **Line and sequence:** pick a **Line** and where it **Runs** (*Before* another job, or *At the end*), then **Move**. For a firmed job you can only change the line; it keeps its start time and slots in by time.
- **Pins:**
  - **Locked:** the job keeps its place when others move, and can't itself be moved until unlocked.
  - **Hold until:** it won't start before this time (for example, waiting for material). **Clear hold** removes it.
  - **Finish by:** Pulaski warns if it is projected to run past this. **Clear finish-by** removes it.
  - Click **Set** to apply.
- **Cancel:** click **Cancel job…**, give a reason, and for a production job optionally tick **Replace it with a job for the quantity not yet made**, then confirm. **Keep it** backs out.

For a job waiting on a **department**, the panel lists the **Lines that could run it**, each with an estimate and whether it **Fits the window** or **Won't fit**; **Run on …** puts it at the end of that line. For a job already taken from a department, **On the floor** lets you move it to another of the department's lines, or **Send back** to the department.

The side panel: a work order

- **Facts:** material, quantity, due date, process and version, origin, notes.
- **Status:** buttons for what it can do next: **Dispatch**, **Un-dispatch**, **Firm**, **Cancel**…
- **Jobs:** its open jobs with their lines and times (click one to open it); overlaps are marked.
- **Place on a line or plan to a department** (planned orders): shows how much isn't on a line yet. Pick a line and a quantity and click **Add job at the end of the line**, or pick a department, set the window (**Window opens** and **Finish by**) and click **Plan to …**.

Planning to a department

Use this when you want a department to finish work in a time window but the floor should pick the line ("Extrusion, this weekend, done before Monday").

1. Plan the job to the department, from the order's side panel or by dragging the order onto the department's row. A department job always needs a **finish-by**.
2. The job waits on the department as a band over its window. Pulaski estimates it on every line below that could run it and warns, without blocking, if even the fastest can't finish in the window, if no line can run it, or if the window has passed.
3. The floor (or you) assigns it to a line: **Run on …** in the side panel, or drag the band or a ghost onto the line. It can move between the department's lines, or go back to the department, whatever its plan status, until it starts.

Pulaski checks each department job on its own; it doesn't share out a department's capacity between jobs.

Dispatching and firming

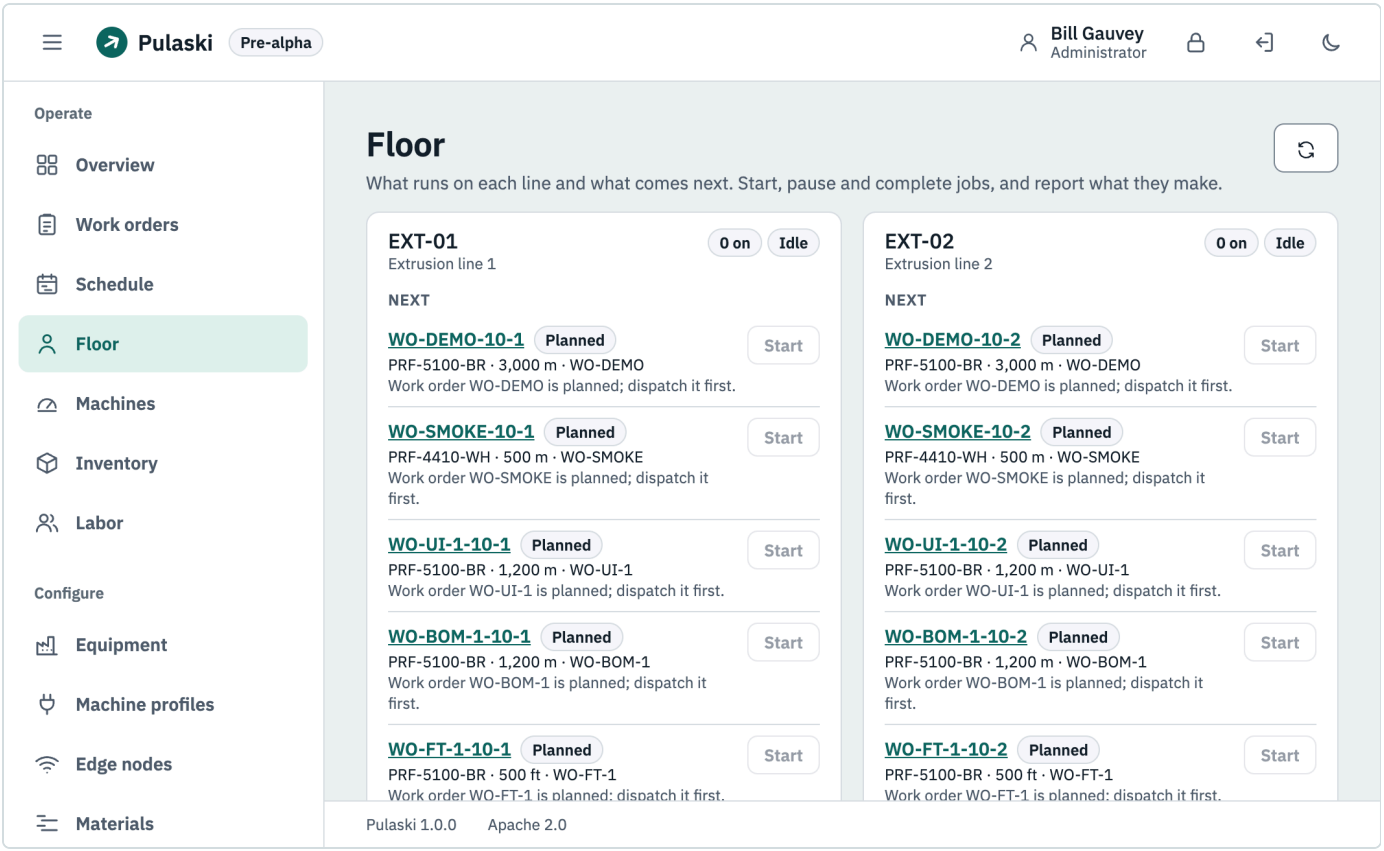
- **Dispatch** publishes the order: Pulaski calculates every job's times on its line right now and saves them as the **baseline**. The floor sees the order, and later you can compare what happened with what was published.
  - Dispatch is refused if the order has no jobs, or if a job's line can no longer run its material.
  - A job waiting on a department is published with its window as the baseline.
  - If an operation's jobs don't add up to the order quantity, dispatch goes ahead but warns, for example *Operation 30 (Pack) plans 1,500 against an order quantity of 2,000.* Each operation is checked on its own, because each makes the whole quantity.
- **Un-dispatch** returns the order and its jobs to planned and clears the baseline. You can do this until the order is firmed.
- **Firm** freezes the plan: no more un-dispatch, and each job is anchored at its baseline start (see Job rules).
- New jobs can only be added to a **planned** order.

Overlaps

Pulaski lays work end to end on each line, so two jobs on a line only overlap when work runs into a **firmed** job's fixed start. When that happens, a red banner appears on every view: *1 overlap: a line can't run these at the same time*, listing each pair with a hint and a **Show** button. The jobs are outlined, and a notice pops up the moment a new overlap appears, whether from your edit, another planner's, or simply time passing. Pulaski never fixes an overlap itself: move or resequence work, or un-firm the order.

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11. The Floor: running jobs and reporting production



Floor: running and upcoming jobs by line. Local demo screen; details vary by site, data and role.

Operate → Floor. Where operators start, pause and complete jobs and report what they make. The schedule says what should run; the floor records what does. Each line shows how many people are on it, and someone set up to clock themselves on sees **Clock on** and **Clock off** on each line (see Labor).

Ideas to know first

- **Runs.** A job runs in one or more **runs**: starting a job opens a run, pausing or completing it ends the run, and resuming opens the next. Good and scrap are reported into a run, and so are the lots the job uses and makes.
- **Only one run at a time on a line.** Pause or complete what is running before starting the next job.
- **When a job can start.** Its work order must be on the floor (dispatched or firmed). If its order was made from a process, the process links must be met: a Pack operation set to start *after Extrude ends* can't start until every Extrude job of the order is completed, plus any lag. Pulaski never starts a job by itself; it tells you when one can start.
- **The rate check.** A run can't hold more than the line could have made in it: its length × the line's **ideal rate** for the material × the site's tolerance (110% unless your administrator changed it). Reporting 1000 m after one hour on a line rated 100 m/h is refused. The check uses the ideal rate from the line's capability, not the planning rate, which is usually lower.
- **The order follows its jobs.** Starting an order's first job firms the order (if it was only dispatched) and puts it **in progress**; completing its last open job **completes** it; reopening a job takes it back to in progress.

The board

Each line is a card. A running line shows its job with the run's length, good quantity against the plan (with a bar), scrap, and how much more the run can hold at the line's rate. Below it, **Next** lists the line's other open jobs in sequence, each with its status and, when it can't start yet, why (*Waiting for operation 10 Extrude to end*) and when (*Ready at 14:30*). Departments with jobs waiting on them have a card of their own. The board refreshes every 15 seconds while no form is open.

Start a job

1. Find the job under **Next** on its line's card.
2. Click **Start** (or **Resume** for a paused job). The button is active only when the job can start and the line is free.
3. For a job waiting on a department, choose the **Line** it runs on next to it, then click **Start**.

If the material changes on a line in automatic changeover mode, Pulaski starts a **changeover** job first (named after the job, such as *WO-5002-10-CO*) and says so. Complete the changeover when the line is ready, then start the job. If no changeover is needed, complete it straight away.

Report production

1. On the running line's card, click **Report**.
2. The **Run** is the open run. Enter the **Good** quantity and the **Scrap** quantity (either can be zero), in the job's unit.
3. For good output, choose where it **goes to** and, optionally, a **Lot number**; left blank, Pulaski numbers it after the job (*WO-5001-10-01*, *-02*…).
4. For scrap, choose a **Scrap reason**.
5. Click **Report**. The card updates with the new totals.

If the report is more than the run can hold, Pulaski refuses it and says how much more fits.

Pause, resume and complete

- **Pause** stops the job for now. Choose why (*End of shift*, *Waiting for material*…) or leave it blank, then click **Pause**. The run ends and the line is free for another job. Click **Resume** later to open a new run.
- **Complete** ends the job. Jobs waiting on it are named when they can now start (*WO-5001-20 on PACK1 can start now.*). If it was the order's last open job, the order is completed.

Report production after the run

Production found after a run has ended (counted at the end of a shift, or a pallet missed) is reported into that run.

1. Open the job's page: click its number on the board, or **Runs and reports** on its inventory page.
2. Click **Report production**.
3. Choose the **Run** it belongs to. A run that has ended is marked *Recorded late*.
4. Enter the quantities as above and click **Report**.

The run's rate check still applies: an hour-long run can't take more than an hour's output. Late reports are allowed until the work order is closed, and show as *late* in the job's reports.

The job page

`jobs/<job>` shows the job's status, planned, good and scrap quantities, its **runs** (line, start, end, length, good, scrap, the most the run can hold, and why it ended) and its **production reports** (when entered, which run, good, scrap, scrap reason, who and any comment). The buttons offer what the job's status allows: **Start** or **Resume**, **Pause**, **Complete**, **Reopen** for a completed job whose order isn't closed, and **Report production**. **Materials and lots** opens the job's inventory page.

Materials made in batches

A material with a **batch size** (set on its material page) is planned and reported in whole batches.

- **Orders and jobs are whole batches.** With a batch size of 50 kg, an order for 125 kg is refused (*125 kg is 2.5 batches of 50 kg… Plan 100 or 150 kg*). A process that shares an order between lines gives each line whole batches.
- **Report batch by batch.** The report form lists **Good batches**, each prefilled with the batch size: change a weight if it differs, and click **Add batch** for each more. Rejected batches go under **Rejected batches**, with a scrap reason. Each good batch becomes its own lot.

- **How close a batch must be.** Under the **near** rule a batch may differ from the batch size by the tolerance (2% by default: 49 to 51 kg for 50 kg), so 49.3 kg is a batch and 40 kg isn't. Under the **exact** rule, the default for finished goods, every batch is exactly the batch size.

A partial batch (the line went down)

A batch cut short is an exception, so it is reported only as the line stops:

1. Click **Pause** (or **Complete**, if the job ends there).
2. Choose **why** it stopped: a partial batch needs a reason.
3. Tick **A batch was in progress**, enter its **Quantity** and where it **Goes to**.
4. Click **Pause** or **Complete**.

The partial batch goes into a lot of its own. Under the near rule it is **accepted**: it will be used. Under the exact rule it goes **on hold until approved**, because a finished good can't be sold as a partial; the floor shows *A partial batch waits for approval*. Each run can have one partial batch. If you forgot it, open the job's page and report it into that run: the form offers **Partial batch** for a run that ended with a reason.

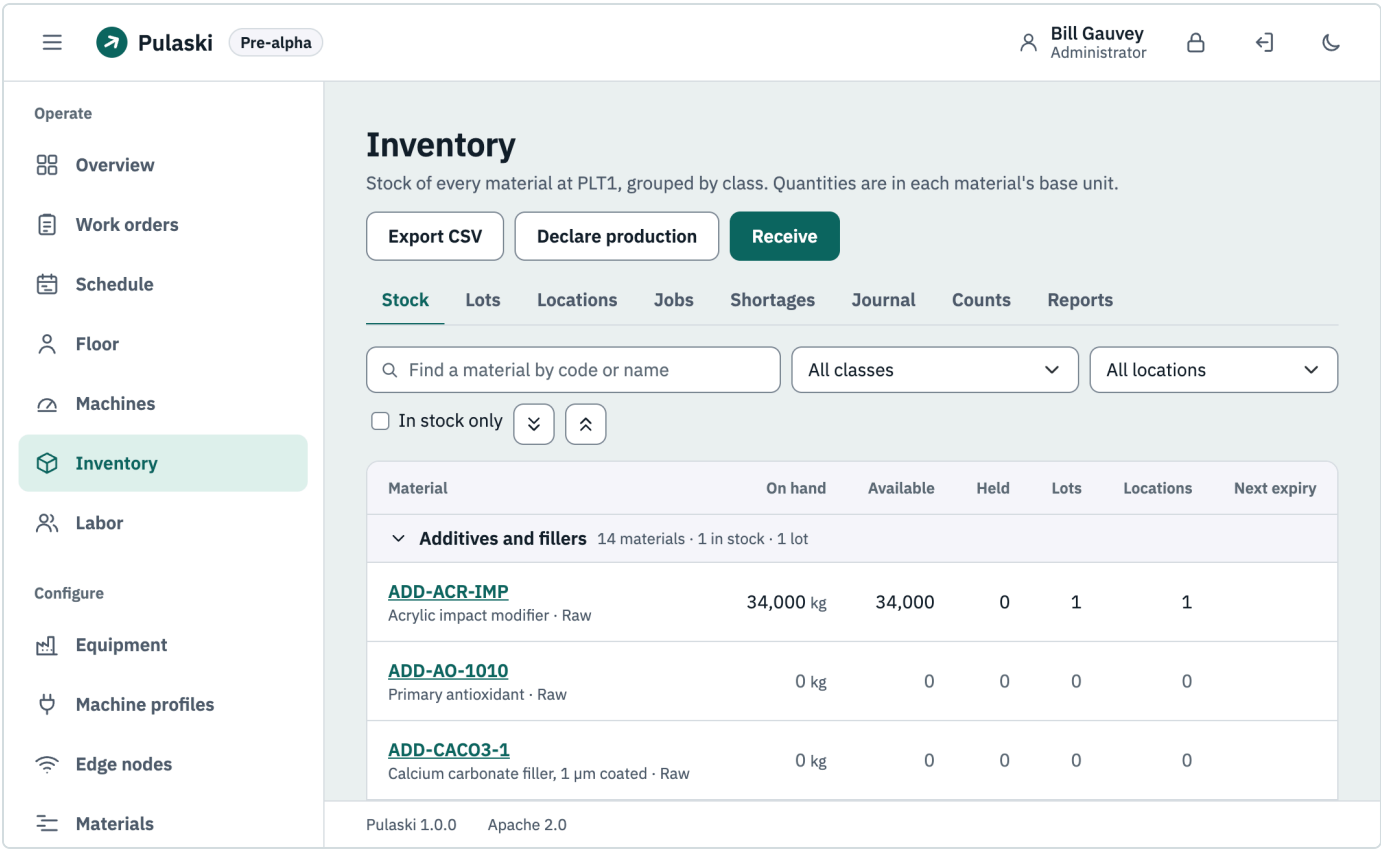
Approve or reject it on the job's page, under **Partial batches**: **Approve** releases the lot; **Reject** asks for a scrap reason, writes the lot off and moves its quantity from good to scrap. Both record who decided and when. Only a Supervisor or Quality user at the site can decide, and never the person who reported it.

In detail

- **Statuses.** A production job starts from *dispatched* or *firmed* (starting firms a dispatched order with all its jobs, since execution freezes the plan), and resumes from *paused*. Non-production jobs (cleaning, maintenance, trial, changeover) also start from *planned*. A completed job can be reopened to *paused* until its order is closed; closing the order closes its completed jobs.
- **The order's status** can't be set to in progress or completed by hand any more: it follows its jobs. Closing a completed order is still done on the Work orders screen.
- **The rate check** uses the capability in effect when the run started, converted into the job's unit, and counts good plus scrap across the whole run, including late reports. If the line had no capability for the material, or its rate unit doesn't convert, the report is recorded with a warning that it wasn't checked. The tolerance is set per site, between 100% and 1000%, through `PUT /api/execution/sites/<site>/policy`; there is no screen for it yet.
- **When a report counts.** A report into the open run counts at the moment it is entered; one into an ended run counts at the end of that run. Both keep the time they were entered.
- **Consumption** needs a run too: the job's open run, or the run chosen (see Record what a job used and made). Using more of a planned component than the plan allows is recorded with a warning. The plan's quantity is scaled up when the job makes more than planned, and gets the site's consumption tolerance (also 110% by default).
- **The changeover check** compares the job's material with the last job that ran on the line. If that was a production job of another material, a changeover job is started first. In *scheduler* mode, and on a line that hasn't run anything yet, no changeover is added.
- **Process links.** *After start* is met once any job of the upstream operation has started; *after end* once all of them are completed. Each is plus its lag. Cancelled upstream jobs don't count.
- **Events:** `pulaski.job.started.v1`, `…paused.v1`, `…completed.v1`, `…reopened.v1` and `…production_reported.v1`; the counts are kept in `ts.production_count`.
- **Batches in detail.** Batch weights are in the material's base unit. A partial batch is more than 0 and less than a batch (under the near rule, less than the size minus the tolerance). The rate check counts it like any output. Under the exact rule a partial batch needs a location that keeps lots apart (not a silo or tank where lots mix). The lot is held with the reason *Partial batch awaiting approval*. When a job of a batch material is cancelled and replaced, what is left is rounded **up** to whole batches. Partial batches are recorded as `pulaski.partial_batch.reported.v1`, `…approved.v1` and `…rejected.v1`.
- **Concurrency.** Start, pause, complete and reopen check the job hasn't changed since the screen read it; reports don't, so several people can report into a busy line, protected by the rate check under a lock.

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12. Inventory



Inventory: stock by material, location and status. Local demo screen; details vary by site, data and role.

Inventory (in the side menu under *Operate*) shows what material you have, where it is and what happened to it. It has eight tabs: **Stock**, **Lots**, **Locations**, **Jobs**, **Shortages**, **Journal**, **Counts** and **Reports**. If your company has more than one site, pick the site at the top right; Pulaski remembers it while you move between the tabs.

Ideas to know first

- **Quantities are in the material's base unit.** You can enter a quantity in any unit that converts (pounds for a resin stocked in kilograms, for example); Pulaski converts it and keeps the base unit. If a unit doesn't convert you'll see *"… doesn't convert to …"*: add a conversion.
- **Stock can't go below zero.** Pulaski refuses to take out more than a lot holds and tells you how much is left.
- **Statuses.** *Available* lots can be used. *On hold* (waiting for a decision) and *quarantine* (kept apart, suspected bad) lots count as on hand but can't be consumed until released. A lot taken down to zero becomes *consumed* or *scrapped*; it stays on record for genealogy.
- **Lot-tracked or not.** For a lot-tracked material every receipt is its own numbered lot. For one that isn't, such as end caps, there is one **balance** at each location (shown as *Balance at RM-STORE*, with no lot number); receipts add to it.
- **Expiry and shelf life.** A material can have a **shelf life** (set on its inventory page): a new lot received or produced without an expiry date expires that many days later. An **expired** lot is marked in amber; it can't be consumed, shipped or merged (into another lot or a silo's balance). It can still be moved, held, written off or counted. If quality extends it, use **Correct…** on the lot to set the new expiry, with a comment saying why.
- **One material at a time.** A location marked **One material at a time** refuses another material, whether received, moved, split there or produced, until the quantity of the material already there is 0.
- **Locations where lots mix.** A silo or tank marked **Lots are indistinguishable** on the Equipment page keeps one balance of each material, whatever it is. Everything that arrives joins it: receipts, moves, splits sent there and production. A receipt with a lot number is kept as that lot first, then merged in, so a trace still leads back to the supplier's lot; with no number it goes

straight into the balance. Only available stock can join an available balance (release a held lot first), and the balance takes the earliest expiry of what joins it. A confirmation says when a lot has joined a balance. A balance that is moved somewhere else stays a balance there: mixed stock stays mixed.

Find what you have: the Stock tab

1. Open **Inventory**. Every active material is listed, grouped by **material class** (*Resins*, *Extruded profiles*…). Each class header shows how many of its materials are in stock.
2. Type in **Find a material** to filter by code or name, or pick a class.
3. Pick a **location** to see only the stock held there.
4. Tick **In stock only** to hide materials with nothing on hand.
5. Click a group header to fold it; use the two buttons at the end of the toolbar to unfold or fold them all.
6. Click a material's code to open it.

Each row shows **On hand** (everything not used up), **Available**, **Held** (on hold or in quarantine, in amber), the number of **lots** and **locations**, and the **next expiry** (amber when it is within 14 days or past).

A material: lots, locations and where used

The material page shows tiles for on hand, available, held, lots and locations, and its **shelf life** in days: click **Change** to set it, or leave it blank for none (lots already on hand keep their expiry). Then:

- **Lots on hand**, with status, quantity, location, supplier lot, expiry and where each came from. **Used-up lots** at the top lists the consumed ones.
- **By location**: the balance at each location. Click one to list its lots.
- **Where used**: every released or draft **BOM** that lists the material as a component (with the quantity per base quantity and the operation that consumes it), and every **open job** planned to consume it, with how much it has consumed so far. **Made from** links to the material's own BOMs.

Find a lot: the Lots tab

1. Type a **lot number**, a **supplier lot** number or a material code, and/or choose a material, location, status or expiry window (*Expired*, *within 7, 30 or 90 days*).
2. Click **Search**. Without a status, only lots with stock on hand are found; choose *Consumed* or *Scrapped* to find used-up lots.
3. Click a lot to open it. At most 500 lots are listed; narrow the search if you hit that.

Receive a lot

1. Click **Receive** (on the Stock tab, or on a material's page, where the material is filled in).
2. Choose the **material** and the **location** it goes to.
3. For a lot-tracked material, enter the **lot number** (required, and unique for that material in the site; optional where lots mix), and if you have them the **supplier lot** and **expiry date**.
4. Enter the **quantity** and its **unit**, and an optional comment.
5. Click **Receive**. A green bar confirms it, with a link to the lot.

A lot's page

The lot page shows its facts (status, quantity, location, where it came from, supplier lot, expiry), its **genealogy** and its **history**: every movement with the time, what happened, the change, the quantity after it, where, the job, the reason and comment, and who did it. The buttons at the top act on the lot:

Button	Does	Needs
Consume	Records that a job used some of the lot.	An available lot with stock. Choose the job; jobs planned to use this material come first, marked *planned*.
Move	Moves the whole lot to another location in the site. For a balance, enter how much to move; it joins the balance there.	A lot on hand.
Write up	Adds stock that was found or miscounted. A used-up lot comes back as available.	A quantity and an adjustment reason.
Write off	Removes stock: damage, loss, a count correction. Leave the quantity blank to write off the whole lot. At zero the lot becomes *scrapped*.	An adjustment reason.
Split	Splits part of the lot into a new lot with the same material, status, supplier lot and expiry, here or at another location. A blank number becomes *R-24-0917-1*, *-2*…	Less than the whole lot (to move all of it, use **Move**).
Merge…	Takes other lots into this one: tick them in the list. They go to zero (*merged into*), and this lot takes the earliest expiry of them all.	Lots of the same material, at the same location, in the same status.
Ship	Sends stock out of the site, for example to a customer or another plant. At zero the lot is *shipped*.	An available lot, and a delivery or shipment number.
Hold… / **Change hold…**	Puts the lot **on hold** or in **quarantine**, or switches between the two.	A hold reason.
Release	Makes a held or quarantined lot available again.	—
Correct…	Fixes the lot number, supplier lot or expiry, for example after a typo on receipt.	A comment saying why. A balance's number can't change.
Reverse (on the latest row of the history)	Undoes the lot's latest movement: a receipt, write-up, write-off, move, hold or release, or shipment. The lot goes back to how it was; a reversed receipt of a whole lot makes it *cancelled*.	An adjustment reason. Only the latest movement, once.

Some reasons ask for a comment; the form says *(required)* when they do.

Record what a job used and made: the Jobs tab

Until Pulaski records production from the machines, operators declare it here. Material is used and made while a job runs, so everything goes into one of the job's runs (see The Floor).

1. Open the **Jobs** tab and click a job (search by job, order or material). Production jobs that are running, paused, or completed but not yet closed are listed; start a job on the Floor first.
2. The job page shows the **components** planned for it (from its order's BOM) with how much has been consumed and how much is still to use, and the **lots it made**.
3. To record material used: click **Consume a lot**, choose the material (planned ones are marked), choose the lot, check the **Run** (the open run is chosen; pick an ended one to record use found afterwards), enter the quantity and click **Record consumption**. Using more than the job's plan allows is recorded with a warning to check it.
4. To record output: click **Declare production**, check the **Run**, choose the location the output goes to, enter the quantity (the job's unit is filled in) and click **Record production**. Leave **Lot number** blank and Pulaski numbers it after the job: *WO-1001-10-01*, *-02*… Enter an existing number of a lot this job made at the same place to add to it.

You can also consume from a lot's own page, and declare production from the Stock tab (choose the job there). Declaring production here is the same as reporting good output on the Floor, with the same rate check; report scrap on the Floor or the job's page.

Trace a lot

On a lot's page, **Genealogy** first shows the lots it was **made from** and the lots made from it (**used in**). Click **Trace back** to see everything it was made from, all the way up, or **Trace forward** to see everything made from it, all the way down. Each lot is indented under the one it was reached from; click any lot to open it.

Check for shortages

The **Shortages** tab answers *"will we have the material for the jobs coming up?"*.

1. Open **Inventory** → **Shortages**. It lists every component the site's open jobs still need, short ones first.
2. Narrow it to a line or department (and everything under it) with the equipment picker, or to jobs needed soon: the next 24 hours, 3, 7 or 14 days.
3. Clear **Short only** to see covered components too.
4. Click a component to see each job that needs it, in the order they are needed: what it still needs, the running total, and whether it is **covered** or how much is short.
5. Click a job to open it, or **stock** to open the component.

Each row shows **Needed** (what the jobs still need: planned less consumed), **Available**, **Held** (on hold or quarantine, in amber: releasing it might cover the gap), **Short** (in red), and the **first short** job with when it is needed.

In detail

- Jobs take stock in the order they are needed: their published start, else their hold-until time, then job number; jobs with neither come last. The live projected time isn't used yet, so a job that has slipped is ordered by its published time.
- With a horizon, jobs needed after it, and jobs with no time at all, are left out.
- Only **available** stock counts. Incoming deliveries and purchase orders aren't counted, and nothing is suggested: ordering stays in the ERP.
- A component planned in a unit that doesn't convert to its base unit is left out, with an amber note naming the job.

The movement journal

The **Journal** tab lists every movement in the site, newest first: receipts, write-ups, write-offs, moves, holds, consumption and production.

1. Open **Inventory** → **Journal**. It shows the last 7 days.
2. Change the **From** and **to** dates (whole days at the site; at most a year), or filter by movement, material, location, lot or supplier lot number, and **Who** (part of the person's name).
3. Click **Search**. Click **Show older** at the bottom for the next 200.
4. Click a lot, a material or a job to open it.

A material's and a job's pages have a **Movements** button that opens the journal filtered to them.

Count a location

A **count** checks what is really at a location and puts the records right.

1. Open **Inventory** → **Counts**, choose the **location**, add a note if you like (*month end*), and click **Start a count**. Pulaski takes a snapshot of every lot there and opens the count.
2. Count each lot and type what you find in **Counted** (in the lot's unit; enter **0** for a lot that isn't there). The **Difference** shows how far it is from what the lot holds now.
3. Click **Save counts** as you go; you can come back and carry on later.
4. When every lot is counted, choose the **reason** (normally *Count correction*) and click **Post**. Each lot that differs is written up or written off, with the count number as its reference. Lots that match aren't touched.
5. **Cancel count** drops it without changing anything.

In detail

- If a lot moves after you counted it (someone consumed or moved it), posting is refused for it, marked *changed since counted*: count it again. A lot that moved after the count started but before you counted it is fine: the count is what is there.
- **Now** turns amber when a lot changed since the count started.
- Stock you find that isn't in the count is received separately. Count numbers are *CNT-* and the date the count started (UTC), then a number.

Reports and exports

Inventory → **Reports** has two reports:

- **Stock on a date:** each material's on-hand quantity at the end of the day you choose. It is worked out from every lot's last movement up to then, so older dates take longer. Held stock is included: a lot's status at that time isn't reconstructed.
- **Aging:** on-hand stock by how long ago each lot was received or made: up to 30 days, 31–90, 91–180 and over 180 (in amber). Balances of materials that aren't lot-tracked aren't aged, and a split lot counts from the day it was split.

Export CSV downloads a spreadsheet file from the **Stock** tab (what is on screen, with its filters), the **Lots** tab (every matching lot, up to 10,000), the **Journal** (every matching movement, up to 10,000) and both reports. Lot search shows 200 at a time: click **Show more** for the next.

Reason codes

Configure → **Reasons** lists the reasons people choose, one tab per kind: inventory adjustments, holds, scrap, run ends, **utilization** (what a machine's time counts as), and kinds later screens will use.

1. To add one, type a **code** (letters, digits, `-`, `_` or `.`), a **name**, choose **Every site** or one site, tick **Needs a comment** if people must explain, and click **Add**.
2. To change one, edit its name, comment setting, order or **Active** box, then click **Save** on its row (**Undo** throws the change away).
3. A code and its kind can't change once created, because history refers to them. To stop using a reason, untick **Active**.
4. A **utilization** reason also has a **Class**: *Runtime* (producing or running slow), *Downtime* (an unplanned stop, counted against availability) or *Planned (not in OEE)* (setup, teardown, cleaning, planned maintenance, breaks). Choose it when you add the reason; you can change it later on its row and **Save**.

Locations

The **Locations** tab lists every piece of equipment that holds inventory, with how many lots and materials are there. **Stock** and **Lots** open those lists for that location. To add a location, add equipment (for example a storage zone) on the Equipment page and tick **Holds inventory**.

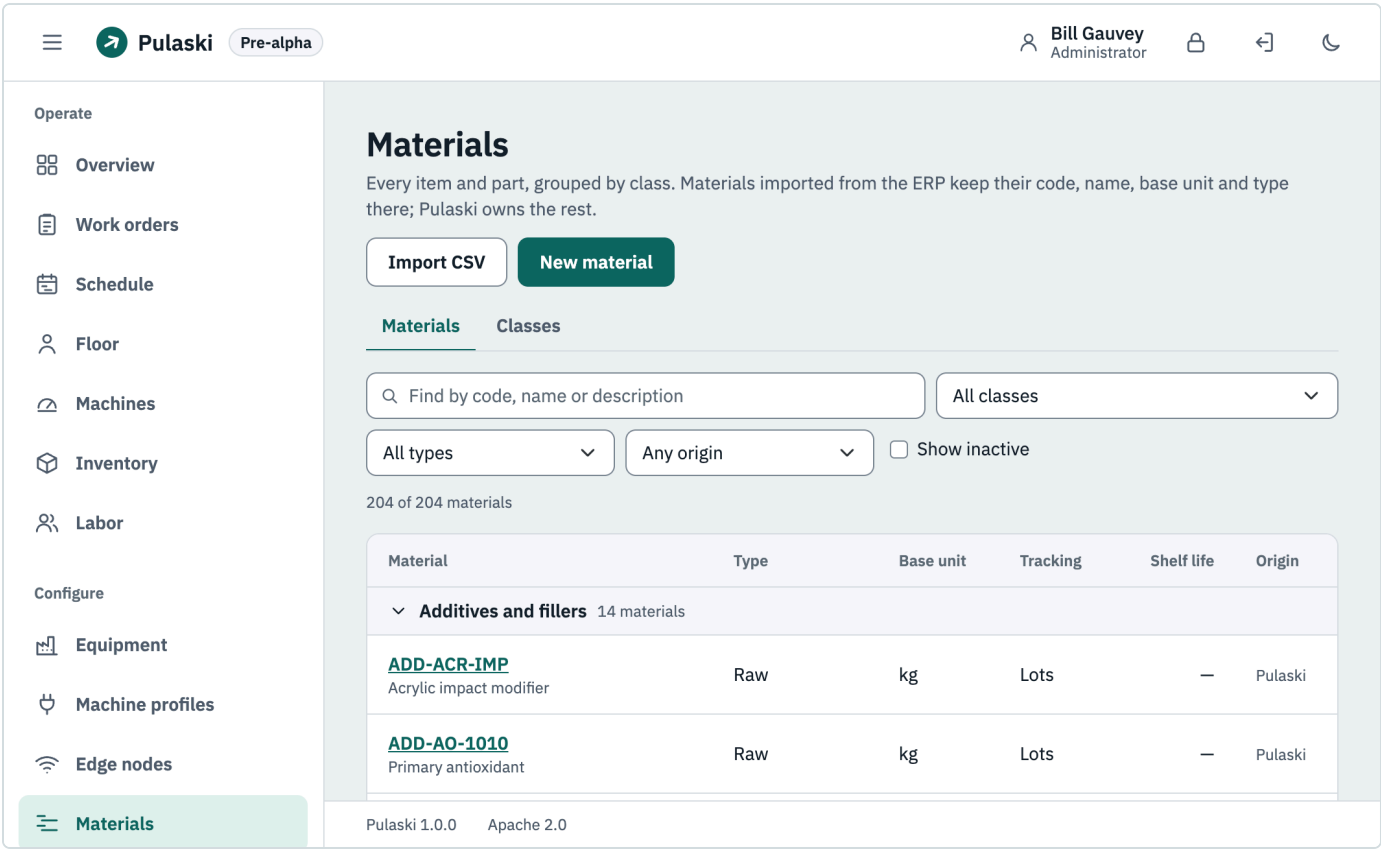
In detail

- **One site per lot.** Moves stay within the lot's site; consumption needs a job in the same site.
- **Concurrency.** Write up, write off, move, hold and release check that the lot hasn't changed since you opened it; if it has you'll see *"changed by someone else"*: reload and try again. Consumption and production don't check, so several people can declare against a busy line; the no-negative-stock rule still protects them.
- **Substitutions.** Consuming a material the job has no planned component of is allowed, and recorded as a **substitution** (amber on the job page and in the confirmation).
- **Genealogy is per job for now.** Every lot consumed into a job is recorded as a parent of every lot the job made, whichever came first. That can link more than really mixed (a resin lot used in the first hour shows as a parent of the last pallet). It narrows to time windows once Pulaski records production from the machines.
- **Times** are the moment you click; you can't back-date a movement. A movement recorded into a run that has ended keeps the time you entered it and names the run it belongs to.

- **What the ERP sees.** Each movement is an event in Pulaski's event log (`pulaski.lot.received.v1` , `…consumed.v1` and so on), ready to be sent to the ERP once the connection is built.

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13. Materials



Materials: search, filtering and material setup. Local demo screen; details vary by site, data and role.

Configure → Materials. Every item and part Pulaski knows about: what you make, what you make it from, and what you pack it in. The page has two tabs, **Materials** and **Classes**.

Ideas to know first

- **Where a material comes from.** A material is either created here (origin **Pulaski**) or **imported from the ERP** (its origin chip shows the ERP's name, such as **ERP1**). For an imported material the ERP owns the **code, name, base unit and type**: you change those in the ERP, and the next import brings them in. Pulaski owns everything else: description, class, lot tracking, shelf life and whether it is active.
- **Type** is a label from the item master, such as the ERP's MRP class: *raw*, *intermediate*, *finished*, *packaging*, *consumable* or *rework*. It is for finding and grouping; no rule depends on it.
- **Classes** group similar materials, such as *Extruded profiles*, so a process or a changeover rule can name a class instead of each material. Classes are one level deep: a class has no parent class.
- **In use** means something refers to the material: a lot, an order, a job, a BOM, a capability or rate, a process or a changeover rule. Some fields are fixed once it is in use (below).

Find a material: the Materials tab

1. Open **Configure → Materials**. Materials are listed by class, with *No class* last. Click a class heading to fold it.
2. Narrow the list: type in **Find by code, name or description**, or pick a class, a type or an origin (**Pulaski** or **ERP**).
3. Inactive materials are hidden; tick **Show inactive** to see them.
4. Click a material's code to open its page.

Each row shows the type, base unit, tracking (**Lots** for numbered lots, **Balance** for one quantity per location), shelf life and origin.

Create a material

1. On the Materials tab, click **New material**.
2. Fill in **Code** (the item or part number, unique) and **Name**, both required.
3. Choose the **Type**, the **Class** (optional; only active classes are offered) and the **Base unit**, the unit stock is counted in (required).
4. Optionally enter a **Shelf life** in days, untick **Lot-tracked** for a material kept as one balance per location, and add a **Description**.
5. Click **Create and open**. The new material's page opens.

Materials from the ERP don't use this form: they arrive through the import (see *In detail*).

The material page

The left side lists the material's properties, grouped into *Identity*, *Classification*, *Stock* and *Status*. Edit one the same way as equipment: text saves when you press **Enter** or leave the field, and a choice saves as soon as you pick it. Click a property to read what it means and, when it is greyed out, why it can't change.

The rest of the page shows what the material is connected to:

- **What uses it:** stock on hand (with the number of lots), open work orders, open jobs, released BOMs that use it as a component, and the lines that can run it. Click the stock tile, or **Stock** at the top, for its inventory.
- **Where it can run:** each line with a capability for it, and its planning rate; see below.
- **How it is made:** the processes that make it (**default** marks the one order creation picks first) and its BOMs.
- **Where it is used:** the BOMs that list it as a component, and how many open jobs plan to use it.
- **Its own conversions,** such as *1 m = 1.2 kg*. Click **Conversions** at the top to add one on the Units page.

Where it can run: capabilities and rates

The **Where it can run** table lists each line that can run the material: its ideal rate, the **planning rate** Pulaski schedules with and where that comes from (*calculated* or *modified*; point at the word to see who set a modified rate and why), the standard changeover and crew, and the days it applies. ★ marks a preferred line. A capability that has ended is hidden; tick **Show ended** to see it in grey.

Add a line

1. Click **Add line**.
2. Choose the **Line**. Only active lines that run jobs are offered; a department gets its lines' capabilities.
3. Enter the **Ideal rate, per hour**, the best the line can do, and its **Rate unit**. The unit must convert to the material's base unit; for metres of profile on a line rated in kilograms, add a conversion first.
4. Optionally enter the **Standard changeover** in minutes (used when no changeover rule matches), the **Crew size**, and tick **Preferred line**.
5. **Starts** is today unless you pick another day. **Ends on** is optional: leave it empty for a capability with no end.
6. Click **Add**. If the line has no planning rate for the material yet, the ideal rate becomes it, and a note says so.

Change or end a capability

- Click **Change** on its row, edit the values and click **Save**. A new ideal rate also updates the calculated planning rate; a modified rate stays as it is, and the note says so.
- Click **End now** when the line can no longer run the material. The capability stays in the list as *Ended*. If open jobs for the material are on that line, a note says how many: they stay, but can't be moved to that line or dispatched on it until it has a capability again.

Set the planning rate yourself

1. Click **Rate** on the line's row. The form shows the calculated rate and what it is based on.

2. Enter the **Modified rate** and a note saying why (for example *trial on new die*), then click **Set modified rate**. Pulaski now plans with your rate.
3. To go back, click **Rate** again, then **Back to calculated**. Whatever is in the note box is kept as the reason.

In detail

- **Periods.** A capability applies from the start of its first day to the start of its end day, in the line's site time zone. One that starts today starts the moment you add it, so it can follow straight on from one you ended earlier today. Two periods for the same material on the same line can't overlap: end or change the other one first.
- **Nothing is deleted.** Ending sets the end to now. A capability that hasn't started yet can't be ended; change its dates instead.
- **The planning rate** is one per line and material, whatever the capability periods. Pulaski calculates it from the ideal rate for now (later from run history; then the ideal rate no longer changes it). You can't change a capability's rate unit while the planning rate has a modified rate in the old unit: clear the modified rate first.
- **Events:** `pulaski.capability.added.v1`, `…changed.v1`, `…ended.v1`, `pulaski.run_standard.created.v1` and `…recalculated.v1`, besides the existing `…modified.v1` and `…modification_cleared.v1`.

Import materials from a CSV file

Load many materials at once, for example when setting up a new site, from a spreadsheet saved as CSV.

1. On the Materials tab, click **Import CSV**.
2. Prepare the file. Its first line names the columns; the panel shows them all. **code**, **name**, **type** and **base_unit** are required; the rest are optional:

Column	Holds
`code`, `name`	Item number and name.
`type`	`raw`, `intermediate`, `finished`, `packaging`, `consumable` or `rework`.
`base_unit`	A unit code Pulaski has, such as `kg`, `m` or `ea`.
`class`	A class code from the Classes tab.
`description`	Notes.
`lot_tracked`, `active`	`yes` or `no` (also `true` / `false`, `1` / `0`).
`shelf_life_days`	A whole number of days.
`external_system`, `external_key`, `external_version`	For items from the ERP: its name, the item's key there, and an optional version.

3. Click **Choose File** and pick the file. Pulaski checks it straight away and saves nothing yet. Each row shows what would happen: **Create**, **Update** (with the fields that change), **Unchanged**, or **Error** with the reason.
4. If there are errors, fix the file and pick it again. Tick **Only rows with errors** to see just those.
5. When every row is fine, click **Import N materials**. The list below updates.

In detail

- **All or nothing.** If any row has an error, nothing is imported. A check run never saves anything.
- **Which material a row is.** A row with `external_system` and `external_key` is the ERP's item: matched on them, created if new, and its code, name, base unit and type updated (as in the ERP import). A row without them is a Pulaski material, matched on its code. A row whose code belongs to an ERP material is an error: give its external columns or change it in the ERP.
- **Left out or blank.** A column the file doesn't have keeps what each material already has (or the default for a new one: lot-tracked, active, no class). A blank cell in `class`, `description` or `shelf_life_days` clears it; a blank `lot_tracked` or `active` keeps it.

- **The rules are the same as on the material page:** a base unit that is in use can't change, lot tracking can't change while stock is on hand, an inactive class can't be given to a material, and a code or ERP key can appear only once in a file.
- **The file:** comma or semicolon separators, quotes around a value that contains them (or line breaks), column names in any order and case (`Base unit` and `base-unit` both work, as do `item_code` and `uom`). An unknown column stops the import, so a misspelt name isn't ignored. At most 5 MB and 5,000 rows per file.
- Each imported row is recorded in the event log as usual, with the reason *CSV import*. The same import is `POST /api/materials/import/csv` (`?dryRun=true` to check), for scripts.

Batches

A material made in batches, such as a dry blend mixed 50 kg at a time, has a batch size. Set it in the property grid under **Batches**:

1. Enter the **Batch size** in the material's base unit and press **Enter**. Leave it blank for a material made continuously.
2. Check the **Batch rule** Pulaski picked: **exact** for finished goods (every batch exactly the batch size; partial batches need approval), **near** for everything else (batches may differ by the tolerance; partial batches are accepted). You can change it.
3. For the near rule, set the **Batch tolerance** (2% by default).

From then on orders and jobs must be whole batches and production is reported batch by batch (see Materials made in batches). Orders already made keep their quantity; Pulaski names any open order that is no longer whole batches.

What can change, and when

Field	Pulaski material	ERP material
Code	Until it is in use	In the ERP only
Name, type	Always	In the ERP only
Base unit	Until it is in use	In the ERP, until it is in use
Lot-tracked	While none of it is on hand	Same
Description, class, shelf life, active	Always	Same
Batch size, rule, tolerance	Always; open orders keep their quantity	Same

- **Base unit.** Quantities are stored in the base unit, so it can't change once the material is in use, not even from the ERP: an import that tries is refused.
- **Lot-tracked.** Switching between lots and balances needs no stock on hand: use, move out or write off what is there first.
- **Shelf life** applies to lots that arrive from now on; lots already on hand keep their expiry.
- **Switching a material off** (setting **Active** to *false*) is always allowed. Afterwards it can't be received, put on new orders or added to BOMs, but what already uses it carries on. A yellow notice lists what still uses it, for example *a capability on 3 lines*. Pulaski never deletes a material.

Classes

1. Open the **Classes** tab. Each class shows its code, name, how many materials are in it, and whether it is active.
2. **Add a class:** type a **CODE** (up to 30 characters, unique) and a name, then click **Add class**.
3. **Rename or switch off a class:** change its name or untick **Active**. **Save** and **Undo** appear on the row; click **Save**.
4. Click a class's material count to see its materials (inactive ones included).

A class's code can't change and a class is never deleted: switch it off instead. An inactive class isn't offered for new materials, and a material can't be moved into one, but materials already in it stay there.

In detail

- **The ERP import** is `POST /api/materials/import`, one item at a time, with the ERP's name, the item's key in the ERP, an optional version, and the code, name, base unit code and type. The first import of a key creates the material; later ones update the code, name, base unit and type, and change nothing (no new version, no event) when nothing differs. The ERP may rename an item's code even while it is in use, since the ERP's key, not the code, identifies it. The base unit must be a unit Pulaski has. To load many at once, use the CSV import. Adapters for other external sources are coming (see What Pulaski doesn't do yet).
- **Codes** are unique across all materials, whatever their origin; an import whose code is already used by another material is refused.
- **Events.** Each change is recorded in the event log as `pulaski.material.created.v1`, `…updated.v1` or `…imported.v1` (and `pulaski.material_class.created.v1` / `…updated.v1`), with the material before and after.
- **Concurrency.** As elsewhere, if someone else changed the material or class since you opened it, you'll see *"changed by someone else"*: reload and try again.

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14. Users, roles and signing in

Everyone signs in to Pulaski. Your **roles** decide what you **see** and what you can **change**: you see only the screens, and only the sites, your roles allow, and the menu shows nothing else. A role is a set of **permissions**, such as *See inventory*, *Receive lots* or *Plan jobs*. An administrator decides what each role may do and gives each user their roles, each at a **site** (only Administrator is everywhere).

Sign in

1. Open Pulaski. It shows **Sign in**.
2. Type your **User name** and **Password** and click **Sign in**.
3. The first time, or after an administrator reset your password, Pulaski asks you to **Change your password**: type the temporary password, then your new one twice, and click **Change password**.
4. To sign out, click the door at the top right.

Your session lasts up to 12 hours. When it ends, or an administrator switches you off, Pulaski shows **Your session ended. Sign in again to carry on.**

Change your password

1. Click the padlock at the top right.
2. Type your **Current password**, then the **New password** twice.
3. Click **Change password**.

A password needs at least 10 characters, with two of: lower case, upper case, digits, symbols, and must not contain your user name. Five wrong passwords in a row lock you out for 15 minutes; an administrator can unlock you sooner by resetting your password.

What each role can change

A new Pulaski starts with these roles. Your administrator may have changed them or added others: the **Roles** and **Permissions** tabs of **Configure** → **Users** show what each role may do today.

Role	Can change
Administrator	Everything, at every site, including users and roles.
Planner	The schedule (place, sequence, pin, assign and cancel jobs); work orders (create, dispatch, un-dispatch, firm, cancel, close); planning rates; materials.
Supervisor	Start, pause, complete and reopen jobs; report production; record and approve labor; code machine stops; receive, move, ship, hold and release lots; stock counts; approve or reject partial batches.
Operator	Start, pause and complete jobs; report production; code machine stops; receive and move lots; clock themselves on and off.
Quality	Hold and release lots; correct inventory (write up, write off, correct, reverse, split, merge); approve or reject partial batches.
Manager	Reopen jobs; cancel and close work orders; approve labor; reason codes; shift schedules; people and crews; the site's settings.
Auditor	Correct inventory; stock counts. Can't plan or run jobs.
Process engineer	Processes, BOMs and which lines can run each material.
Edge node	Nothing on screen: it is for the user a Pulaski Edge node signs in as, to read its configuration and forward machine data.

Equipment (with machine profiles and edge nodes) and units belong to the Administrator. Someone with two roles can do what either allows.

When you can't change something, Pulaski leaves out the button, or switches the fields off and says *You can view this, but changing it needs permission to …*. If you should be able to, ask an administrator.

What each role can see

You see a screen, and its entry in the menu, only if one of your roles lets you see it at your site. A permission to change something includes seeing it: someone who may start jobs sees the Floor. A new Pulaski starts with these:

Role	Sees
Administrator, **Auditor**, **Manager**	Every screen (Users only for Administrator).
Planner	Overview, Work orders, Schedule, Floor, Machines, Inventory, Materials, Processes and BOMs.
Supervisor	Overview, Work orders (to look at), Floor, Machines, Inventory, Labor.
Operator	Overview, Work orders (to look at), Floor, Machines, Inventory.
Quality	Overview, Floor, Machines, Inventory, Materials.
Process engineer	Equipment, Materials, Processes and BOMs, Units.

- **Start page.** After signing in you land on the Overview, or, if you can't see it, the first screen in the menu you can.
- **A screen you can't see** isn't in the menu, links to it show as plain text and buttons for it are left out. Typing its address shows **Page not found**.
- **No screens at all.** Someone whose roles let them see nothing lands on **Nothing to see here yet**, with a button to sign out. Ask an administrator for a role.
- **Sites.** A role held at one site shows that site's data only: other sites' work orders, lots, lines and schedules aren't listed.

Add a user (administrators)

1. Open **Configure** → **Users** (the **Users** tab) and click **New user**.
2. Fill in **User name** (letters, digits and `. - _ @`), **Name**, an optional **E-mail** and a **Temporary password**.
3. Choose a **Role** and, for every role but Administrator, the **site** they hold it at.
4. Click **Create user**, and tell them the user name and temporary password yourself: Pulaski sends nothing.

Change a user's roles, details or password (administrators)

1. On **Configure** → **Users**, **Users** tab, click **Change** on the user's row.
2. **Details:** change the name or e-mail, or untick **Can sign in** to switch them off; click **Save details**.
3. **Roles:** choose a role (and site) under **Add a role…** and click **Add**; click **Take away** to end one.
4. **Password:** type a **Temporary password** and click **Reset password**. They must change it when they next sign in; this also unlocks them.

Add or change a role (administrators)

1. Open **Configure** → **Users** and the **Roles** tab.
2. Click **New role**, or **Change** on a role's row.
3. For a new role, type a **Code** (lower-case letters, digits and `_`, such as `line_lead`; it can't change later) and a **Name**. Add what it's for if you like.
4. Under **May change**, tick each permission the role should have. They are grouped by area: Planning, Floor, Inventory, Configuration, Administration.
5. Click **Create role** or **Save role**. Then give the role to people on the **Users** tab.

To stop using a role, untick **Active** and save. A role someone holds can't be switched off: give them another role and take this one away first. **Administrator** has every permission and can't be changed.

Give roles permissions, one at a time (administrators)

1. Open **Configure** → **Users** and the **Permissions** tab. Each row is a permission, grouped by area; each column is a role.
2. Tick a box to give the role that permission; untick it to take it away. It is saved straight away and applies to everyone who holds the role.

In detail

- **Viewing permissions** are listed first on the Permissions tab, under *Viewing*: one per screen (Overview, Work orders, Schedule, Floor, Inventory, Equipment, Materials, Processes and BOMs, Units, Reasons, Shifts). A view a role gets with a change permission shows ticked and locked, with what it comes with when you point at it.
- **Permissions belong to Pulaski; roles belong to you.** The list of permissions is fixed by the version of Pulaski you run, and grows when new screens need one. Roles, their names and which permissions they hold are yours to change. Administrator always holds every permission, including new ones.
- **Changes to a role apply at once** to everyone who holds it, on their next click.
- **Every user needs a role.** A new user comes with one, and a user's last role can't be taken away: add another first, or switch them off.
- **Someone stays administrator.** The last active administrator can't lose the role or be switched off, and nobody can switch themselves off.
- **Roles are held at a site.** A Supervisor at PLT1 can start jobs at PLT1 only. Something that belongs to no site, such as a reason for every site or an enterprise-wide process, needs the permission at any site.
- **Partial batches:** whoever reported a partial batch can't approve or reject it; the job page says *You reported it: someone else decides*.
- **History** names who did what by user name, such as *oli.operator* on a lot's history.
- **Changes apply at once.** Switching someone off or taking a role away takes effect on their next click, without waiting for their session to end.
- **Ended, not deleted.** Taking a role away from someone ends it, and a role or permission is switched off rather than deleted; Pulaski keeps who held which role when, and who changed a role.
- **The first administrator** is created when Pulaski starts with no users, from the `Security:BootstrapAdmin` settings, with a temporary password to change at the first sign-in.
- **For integrators:** the API takes a bearer token from `POST /api/auth/sign-in`; every other call needs one, and a call without the permission is refused with 403 and the permission in words.

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15. Shift schedules

Configure → Shifts. When each site works. Planning places jobs only inside shifts, so a job that would run into a weekend waits for Monday's first shift. A site can have several schedules: two shifts on weekdays for packing, around the clock for extrusion.

Ideas to know first

- **A schedule** belongs to a site and uses its time zone. It repeats every so many days: 7 for a weekly schedule, 14 or 28 for a rotation. It has a day the cycle starts on, and shifts, each with a start time, a length and the days of the cycle it runs on.
- **Equipment follows a schedule, inherits one or runs around the clock.** On Equipment, each site, area or line has **Shifts**: *Inherit from parent* (the default), one of its site's schedules, or *Around the clock*. A line uses the nearest setting, its own first, then its area's, then its site's. With nothing set anywhere, it runs around the clock and the schedule marks it **24/7**.
- **Changes apply at once.** Change a schedule, or which one a line follows, and the schedule shows the new times on its next refresh.

Add a shift schedule

1. Open **Configure → Shifts**. With more than one site, choose the site at the top.
2. Click **New schedule**. It starts as one day shift, 06:00 for 8 hours, Monday to Friday.
3. Type a **Code** (such as `2X8`; it can't change later) and a **Name**.
4. Set **Repeats every (days)** and **Cycle starts on**. For a weekly schedule leave 7 and a Monday.
5. For each shift, set its **Shift** code, **Name**, **Starts** time, **Hours** and, if you record labor, the **Crew** that works it, and tick the days it runs. **Add shift** adds one that starts when the last one ends, on the same days; **×** removes one.
6. Check **The next 7 days** below the grid: it lists the shifts as they will fall, and the hours per cycle. Problems, such as two shifts overlapping, show there instead.
7. Click **Create schedule**.

Make a line follow it

1. Open **Configure → Equipment** and select the site, area or line.
2. Under **Working time**, set **Shifts** to the schedule. Everything below it that inherits now follows it too.
3. **Shifts in effect** shows the result. To make one line run around the clock under a site that has shifts, set that line's **Shifts** to *Around the clock*.

Change or switch off a schedule

1. On **Configure → Shifts**, click **Change** on the schedule's row.
2. Change its shifts, cycle or name and click **Save schedule**.
3. To stop using it, untick **Active**. A schedule that equipment follows can't be switched off: give the equipment another setting first. **Followed by** lists that equipment.

In detail

- **A shift belongs to the day it starts on:** a night shift from 22:00 Monday to 06:00 Tuesday is Monday's. Shifts may run past midnight but may not overlap, also from the last day of the cycle into the first.
- **Days in the grid** are weekdays (*Mon*, *Tue*) when the cycle is a whole number of weeks, with the week number (*W2 Mon*) beyond one week, and *Day 1*, *Day 2* otherwise.
- **Daylight saving:** shifts keep their clock times. A shift that would start in the hour the clocks skip starts when they have gone forward; a shift's length is real time, so a night shift is an hour longer or shorter on the night the clocks change.
- **Hours a cycle** is the working time in one cycle, the sum of each shift's length times the days it runs.
- **Calendar exceptions** (holidays, shutdowns) still take time out of the shifts, but have no screen yet.

- **Who:** maintaining schedules, and choosing which one equipment follows, needs permission to maintain shift schedules (the Manager role has it to start with). Seeing them needs permission to see shift schedules, or to see equipment.

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16. Labor, people and crews

Who worked on which line, for how long, and against the standard crew. Supervisors check their crew onto a line at the start of a shift, operators can clock themselves on, and at the end a supervisor approves the shift's labor. Pulaski doesn't do payroll or attendance: export the hours for that.

Ideas to know first

- **People and crews.** Everyone whose time is recorded is a **person**, with an employee number and usually a **crew**. A person needn't sign in to Pulaski; link one to a Pulaski user to let them clock themselves on.
- **The crew on shift.** Each shift in a shift schedule can name its crew, so Pulaski knows who should be on. A rotation is several shifts at the same time on different days, each with its crew.
- **Kinds of time.** *Direct* is running the line, *setup* is getting a job ready (it names the job), *indirect* is time off production such as cleaning (it needs a reason), *training* is training.
- **Jobs change, people don't re-clock.** Direct time on a line counts towards whatever ran on it while the person was there. Time on a line with nothing running is shown as *idle*.
- **A share of their time.** Someone tending two lines is on each at 50%. A person's time adds up to at most 100%.
- **Shifts and approval.** Time belongs to the shift running when it started (or starting within two hours). Once a shift has ended, a supervisor approves it: open time ends when the shift ended, and the shift is locked. After that, a change is a **correction** with a reason.

Set up people and crews

1. Open **Configure** → **People**. With more than one site, choose the site at the top.
2. On **Crews**, type a **Code** (such as `RED`; it can't change later) and a **Name**, and click **Add crew**.
3. On **People**, fill in **Employee no.**, **Name**, the **Crew** and, if they will clock themselves on, the Pulaski **User name** they sign in as. Click **Add person**.
4. In **Configure** → **Shifts**, set the **Crew** of each shift.

Check a crew onto a line

1. Open **Operate** → **Labor**. **On now** shows each line with its shift, crew, the job running and who is on (with the standard crew, when there is one).
2. Click **Check crew in** on the line. The crew on shift is ticked; people already on another line are listed but not ticked.
3. Untick anyone who isn't there, tick anyone else, and click **Check … in**.
4. To add one person for other time, click **Add someone**, choose who, the kind of time, the reason (for indirect) and their share, and click **Clock on**.
5. Click **End** next to someone to end their time, or **Check everyone out** at the end of the shift.

Clock yourself on (operators)

1. Open **Operate** → **Floor**.
2. Click **Clock on** on the line you are working on. Pulaski refuses if you are already on another line with all your time: clock off there first.
3. Click **Clock off** when you leave it.

Review and approve a shift

1. Open **Operate** → **Labor** → **Shift review** and choose the **Day**. Each shift that has labor is a button; one with *approved* is locked.
2. Check the people, times and hours. Click **Change** to fix a start, end or share, or **End** for someone still on.
3. Once the shift has ended, add a note if you like and click **Approve shift**. Anyone still on is clocked off at the end of the shift.

4. To fix an approved shift, click **Correct** (change the times or share) or **Remove**, say why, and click **Save**. The original row stays, struck through; the hours count the correction.

Labor reports

1. Open **Operate** → **Labor** → **Reports**, choose **From** and **to** (production days, up to 93), and click **Show**.
2. **By job** shows run hours, direct and setup hours, the standard (run hours × the line's standard crew size for the material) and **efficiency** (standard ÷ direct hours: under 100% took more people than the standard).
3. **By person** shows each person's hours by kind, and how much of their direct time was idle. **By shift** shows each shift's hours and whether it is approved.
4. **Export CSV** on each table saves it, for payroll or the ERP.

In detail

- **Who:** seeing Labor needs permission to see labor; checking crews in and changing time needs permission to record labor; approving and correcting needs permission to approve labor; People needs permission to maintain people and crews. Clocking yourself on needs permission to clock yourself on, and being set up as a person.
- **One line at a time per person**, and their open time at most 100% of their time in total.
- **Times can't be ahead of now.** Setup needs a job (the job running on the line, when added from **On now**); indirect needs an indirect labor reason.
- **Switching someone off** is refused while they are clocked on.
- **Nothing runs on its own.** Time stays open until someone ends it, checks the line out, or approves the shift.
- **Hours** are elapsed time × share; an open session counts until now.

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17. Machine profiles

Configure → Machine profiles. A machine profile describes how one kind of machine reports itself: the values its state tag takes and the PackML state each one means, and its own stop, alarm and warning codes with the utilization reason each one counts as. Identical machines share a profile; machines from different vendors or of different models each get their own. You give each machine its profile on the Equipment screen.

Machine data arrives through edge nodes: the profile turns what each machine sends into PackML states.

Ideas to know first

- **PackML states.** Pulaski records machine state as the 18 standard PackML states (0 Undefined, 2 Stopped, 4 Idle, 5 Suspended, 6 Execute, 11 Held…). Many controllers send those numbers as they are; some number their states another way. A profile says what each value means.
- **Stop codes.** When a machine stops, it reports a first-out code: the first thing that went wrong. These codes are the vendor's own, so each profile lists them, with their message and category.
- **Utilization reasons.** Each code, and each state, can count as a utilization reason, such as *Jam* (downtime) or *Setup* (planned). A code with no reason shows as uncoded, for an operator to code.

Add a profile

1. Open **Configure → Machine profiles**. With more than one site, choose the site at the top.
2. Type a **Code** (such as `PACKER-A`; it can't change later), a **Name**, and optionally the **Vendor** and **Model**. Click **Add profile**.
3. The profile opens. It starts with PackML's own 18 states.

Import the machine's definitions from a CSV file

1. On the profile, click **Import CSV**.
2. Choose the file. Pulaski checks it and shows what it would do, row by row: *Add*, *Change*, *Switch off* or *Unchanged*, or the errors.
3. If every row is valid, click **Import**. Nothing is saved until every row is valid.

The file's first line names the columns, in any order and case:

Column	Means
`kind`	`state`, `stop_reason`, `alarm` or `warning`. Required.
`value`	The number the machine sends. Required.
`name`	The state's name, or the code's message. Required.
`packml_state`	State rows only: the PackML state it means, as a number (0 to 17) or a name such as `Execute`. Leave it empty on code rows.
`category`	The vendor's grouping, such as *Mechanical*. Optional.
`reason`	The code of the utilization reason it counts as, such as `JAM`. Optional.

Each kind in the file replaces that list: its rows are added or changed, and entries of that kind the file leaves out are switched off. A kind the file doesn't mention stays as it is, so a file of stop reasons only keeps the states. An example file is `docs/samples/machine-profile-example.csv`. **Export CSV** saves the profile's active entries in the same format, to edit and import again.

Change a profile

1. **States**: change a value's **Name**, **PackML state** or **Reason with no stop code** (what time in that state counts as when the machine gives no code, such as Suspended = *Starved*), or untick **Active**, then click **Save** on the row. To add a value, fill in

the row above the table and click **Add state**.

2. **Stop reasons**, **Alarms** and **Warnings**: change the **Message**, **Category** or **Reason**, or untick **Active**, and click **Save**. A stop code without a reason is marked **No reason**. Add a code with the row above the table.
3. **Reasons offered**: tick the utilization reasons operators may choose when they code a stop on this profile's machines, and click **Save offered reasons**. With none ticked, every utilization reason of the site is offered; once you tick some, operators see those and the ones its codes and states map to (marked *mapped*).
4. The name, vendor, model and **Active** box at the top save with **Save**.

One machine is different

If one machine's code means something else than on the others with the same profile:

1. Open the profile's **Machines** tab. It lists the equipment using the profile.
2. Under the machine, choose the **Stop code** and its reason on that machine (or *Leave uncoded*), and click **Override**.
3. Click **Use the profile's** to remove it again.

In detail

- **Who:** seeing profiles needs permission to see equipment; changing them, and the machine data on equipment, needs permission to maintain equipment at the site.
- **A profile belongs to a site**, and maps only to reasons for every site or for that site. Its code is unique in the site.
- **Nothing is deleted.** States and codes are switched off; an unticked reason stays recorded as unticked.
- **Switching a profile off** is refused while active equipment uses it: give those machines another profile first.
- **Two people at once:** every change to a profile's lists counts as a change to the profile, so someone saving after another's change is asked to reload.
- **Errors in a file** name the line: a value that isn't a whole number, an unknown kind, a missing or unknown PackML state, a PackML state on a code row (usually a missing column), a value that appears twice, a name too long, or a reason that isn't an active utilization reason of the site.

18. Edge nodes: collecting machine data

Configure → Edge nodes. A Pulaski Edge node is a small program on plant hardware, near the machines. It reads them through an MQTT broker they publish to in **Sparkplug B**, keeps what it reads in a buffer on its own disk, and forwards it to Pulaski. If the network or Pulaski goes down, it keeps reading and forwards everything once it can, so nothing is lost.

Ideas to know first

- **Connections** are where an edge node reads from: today, an MQTT broker. Pulaski listens as a Sparkplug B host application.
- **Tags** bind a signal to equipment: *this* metric is PACK1's state, *that* one its processed count. A Sparkplug B address reads `group/node/device:metric`, for example `Plant1/Packing/PACK-01:Status/StateCurrent`.
- **What Pulaski does with them.** A state value goes through the equipment's machine profile to its PackML state, and every change of state is kept, with the value the machine sent. The machine's first-out stop code is kept until it runs again. Counter readings become counts, in the job running on the equipment. Losing the signal is recorded as *Undefined*.

Add an edge node

1. Open **Configure → Edge nodes**. With more than one site, choose the site.
2. Type a **Code** (such as `GW-PACK`; it can't change later) and a **Name**, and click **Add edge node**. Its page opens.
3. On **Connections**, give the broker a **Code** and **Name**, its **Broker host** and **Port**, tick **TLS** if it uses it, and optionally the Sparkplug **Group id** (to hear only one group), a **Host id** (Pulaski then announces itself, so nodes know it is listening), a **Client id** and a **User name**. Click **Add connection**.
4. On **Configure → Users**, add a user for the edge (such as `edge-gw-pack`) with the **Edge node** role at the site. Sign in as that user once and set its password, then sign out.
5. On the node's **Setup** tab, copy the configuration into the edge's `appsettings.json` on the edge machine, fill in the user's password (and the broker's, if it has a user name), and start Pulaski Edge.

Bind tags

1. Once the edge runs, open its **Seen** tab: every metric its brokers have announced, with its last value. Type in **Filter addresses** to find one.
2. Click **Bind** next to a metric. Pulaski guesses the role from the name; choose the **Equipment** (only equipment of the site that tracks state is offered) and check the **Role**.
3. For a count, set the **Scale** (how many units one count is) and, if the counter wraps, its **Rollover** value. Click **Bind tag**.
4. To type an address instead, use the same form on the **Tags** tab. To change a tag, edit its address, scale, rollover or **Active** box and click **Save**.

The edge reads its tags every minute, so new and changed ones take effect without restarting it.

Watch it work

- The node's heading shows **Forwarding** while it is heard from, **Quiet** after 2 minutes without anything (offline, or nothing changed), and when it was last heard from and its version.
- **Live** shows each bound machine's PackML state (and the value it sent, where different), since when, its unit mode, latched stop code and speed. Click **Refresh** to update. Its stops are on Machines.
- **Tags** shows each tag's last value.

In detail

- **Who:** seeing edge nodes needs permission to see equipment; adding and changing them, their connections and tags needs permission to maintain equipment at the site. The edge's own user needs only the **Edge node** role and sees no screens.
- **Sending once.** The edge numbers everything it buffers and Pulaski applies each number once, so forwarding again after a failure changes nothing. A new buffer file starts a new numbering, which Pulaski recognises.

- **What the edge forwards:** a tag's value when it changes, when it is lost, and the last value seen when the tag is newly bound; a speed at most every 5 seconds. A connection it hears from without a birth certificate is asked for one (a Sparkplug *rebirth*), as is every node its tags name when it connects, so an idle machine's state is known.
- **Stop codes** are latched: a machine that clears its code once acknowledged keeps it in Pulaski until it runs (Execute) again.
- **Counts** are taken across the rollover value; a counter going back without one is taken as reset to zero. Machine counts go into the run open on the equipment but don't change the job's reported quantities yet: production is still reported on the Floor.
- **Switching a node off** stops Pulaski taking its data and the edge collecting. Switching a tag or connection off stops it being read.
- **Without Pulaski**, the edge starts with the last configuration it kept.
- **A demo machine.** `deploy/simulator/compose.yaml` runs an MQTT broker and an open-source PackML simulator that publishes Sparkplug B as group `Plant1`, node `Packing`, device `PACK-SIM`. `deploy/simulator/packml-simulator-profile.csv` is a machine profile for it (it numbers Idle and Clearing the other way round). Start the machine with `dotnet run --project tools/Pulaski.SparkplugCommand -- spBv1.0/Plant1/DCMD/Packing/PACK-SIM Command/Reset=true Command/Start=true`.

19. Machines: live status and stops

Operate → Machines. What every machine is doing now, as its edge node reports it, and its stops to code.

Ideas to know first

- **A stop** is time a machine isn't executing: it starts when the machine leaves Execute (or loses its signal) and ends when it executes again. The states it passes through on the way (Stopping, Stopped, Resetting…) are one stop.
- **Its reason.** A stop takes the reason its first-out stop code maps to in the machine's profile; with no code, the reason of the first state on its way that has one (Suspended = *Starved*, say). An operator confirms that reason or gives another, and both are kept.
- **Micro-stops** are stops shorter than the machine's micro-stop threshold (5 minutes unless set otherwise on the Equipment screen). They are listed but not coded, and don't count against availability.

The status board

1. Open **Operate → Machines**. With more than one site, choose the site.
2. Each tile is a piece of equipment that tracks state, coloured by what it is doing: **Producing** (green), **Waiting** (starved or blocked), **Held**, **Stopped**, **Ready** (idle, starting, resetting…), and **No signal** or **Not reporting** (hatched).
3. A tile shows the PackML state and for how long, the unit mode when it isn't Production, the stop in progress with its reason (or **uncoded**) and stop code, the job running with its good count so far, and the speed.
4. A line sits first in its row, with the machines below it. A line set to take its state from its machines says so: **Worst of its machines** (series), **Best of its machines** (parallel) or **Its constraint machine**.
5. **… stops to code** on a tile opens that machine's stops. **… quiet** means its edge node hasn't been heard from for 2 minutes.

The board refreshes every 5 seconds.

Code stops

1. Open the **Stops** tab. The heading shows how many are waiting. **Only stops to code** is ticked: stops with no reason, longer than the micro-stop threshold, including one still going on.
2. Choose the **Equipment** and **Period** (8 hours, 24 hours or 7 days) to look at. Untick **Only stops to code** to see every stop, with its reason; tick **Show micro-stops** to see those too.
3. Click **Code** on a stop. Choose the **Reason**, add a **Comment** if you like (some reasons need one), and click **Save**. Where the stop code mapped a reason, **Confirm …** accepts it in one click; until then the list shows it as *to confirm*.
4. **Change** on a coded stop gives it another reason. The list shows who coded it, their comment, and the reason the code gave when the two differ.

In detail

- **Who:** seeing Machines needs permission to see machines; coding stops needs permission to code machine stops (Operators and Supervisors have both).
- **Which reasons are offered:** with reasons ticked on the machine's profile (**Reasons offered**), those and the ones its codes and states map to; with none ticked, or no profile, every utilization reason of the site.
- **Stop code:** the first one the machine reports during the stop is the stop's, even if the machine reports another later. A machine's own override (on the profile's **Machines** tab) beats the profile.
- **Unit mode:** a stop keeps the unit mode it began in; one in Maintenance or Manual is shown with that mode.
- **Counts on the board** are the machine's processed less defective counts in the run open on it.
- **Each coding is recorded** as an event, with the reason before and after and whether it confirmed the mapped one.

20. Reference

Work order statuses

Status	Meaning	Can go to
Planned	Being arranged; jobs move freely.	Dispatched, Cancelled
Dispatched	Published to the floor with a baseline.	Planned (un-dispatch), Firmed, Cancelled
Firmed	Plan frozen; jobs anchored at their start.	In progress (when its first job starts), Cancelled
In progress	Being made: its first job has started.	Completed (when its last open job completes)
Completed	All its jobs are done; production can still be reported into their runs.	In progress (a job is reopened), Closed
Closed	Done and closed: its jobs are closed and take no more reports.	—
Cancelled	Not going ahead. Cancel its open jobs first.	—

Job rules

Job status	Change line	Change place in sequence	Cancel
Planned	Yes	Yes	Yes
Dispatched	No: un-dispatch the order first	No	Yes
Firmed	Yes, to a line that can run it; it keeps its time	No	Yes
Running, paused	No	No	Yes, optionally replaced
Completed, closed	No	No	No

- A **locked** job can't be moved at all until it is unlocked.
- A **firmed** job is placed at its firmed start (or now, if that has passed) whatever runs before it.
- **Cancel and replace** creates `<job number>–R` for the quantity not yet made, in the same slot.
- Non-production jobs (changeover, cleaning, maintenance, trial) belong to no order and can always be moved unless locked.
- In progress and completed follow the order's jobs: they are set by starting and completing jobs on the Floor, not by hand.

Schedule warnings

Warnings never block anything; they tell you something needs a look.

Warning	Means	What to do
Late	Projected to finish after the order's due date (or a department window closes after it).	Resequence, move to a faster line, or agree a new date.
No run standard	There is no rate for this material on this line, so its length can't be calculated.	Add a capability for the line on the material's page; it sets the rate.
Uom mismatch	The job's unit and the line's rate unit don't convert for this material, so its length is unknown.	Add a conversion for the material.
Beyond calendar	The job runs more than a month past the schedule's horizon, so that part is placed around the clock.	Nothing; it is far ahead.
No shift calendar	No shift schedule covers the line: times assume 24/7.	Choose a schedule for the line or something above it on Equipment, or set it to *Around the clock* on purpose.

Warning	Means	What to do
Missing changeover	In *scheduler* changeover mode: the product changes with no changeover job between.	Add a changeover job.
No duration	A non-production job has no planned length.	Give it a duration.
Firmed conflict	Work runs into a firmed job's anchored start (an overlap).	See Overlaps.
Missed finish by	Projected to finish after its finish-by time.	Resequence or move it.
Window too short	Even the fastest line can't finish the department job in its window.	Widen the window or split the job.
Window passed	The window has closed and no line has taken the job.	Assign it or re-plan it.
No runner	No line in the department can run this material any more.	Add a capability, or plan elsewhere.

Tile meanings

Tile	Counts open orders that…
Starting today	have their first job projected to start today and not running yet.
Finishing today	have their last job projected to finish today.
On track	are on a line, projected to finish by the due date, and starting when published.
Behind	are projected to finish after the due date, or to start more than 15 minutes after the published baseline.
Need attention	have a job with a warning or an overlap.
Not on a line	are planned with quantity not yet on a line or department.

An order can count in several tiles (starting today *and* behind, for example).

Changeovers

Between two production jobs on different materials, a line needs a changeover. Pulaski sizes it from the plant's changeover rules (for example *white to grey: 30 minutes*, *grey to white: 2 hours*), or from the line's standard changeover for the incoming material. By default Pulaski inserts these automatically as placeholders and recalculates them as you resequence. A plant can instead choose to have planners place changeover jobs themselves; Pulaski then warns where one is missing. Your administrator sets which.

21. What Pulaski doesn't do yet

Pulaski is pre-alpha. These are known gaps, so you don't spend time looking for them:

- **No screens yet for** calendar exceptions (holidays, shutdowns) or the changeover setting. Your administrator sets these up.
- **Shifts are for planning and labor only:** a one-off change to a single shift (an extra Saturday), and counts or OEE by shift aren't built yet.
- **Labor:** no badges or shop-floor terminals, no qualifications, no breaks, and no import of people from HR or the ERP. A person works only at their own site. On **Shift review** only a session's times and share can be changed, not its job or kind (end it and add a new one instead). The Overview's *Labor to approve* is still a placeholder.
- **No adapters yet** for other external sources or for label printing: materials come from the screen, a CSV file, or the ERP one item at a time through the API. The CSV import doesn't take capabilities, doesn't create missing classes, and there is no template file to download (the panel shows the header line).
- **Planning rates are calculated from the ideal rate only;** calculating them from run history isn't built yet.
- **Machine data:** there is no OEE, utilization report or stop Pareto yet, and stops can't be split, merged or entered by hand. Machine counts don't replace production reports. The Overview still shows no machine state. Only Sparkplug B is read (no OPC UA or EtherNet/IP yet), alarms, warnings and process values aren't recorded, and Pulaski Edge runs as a console program or container, not yet as a Windows service.
- **Production is reported by hand.** Counts from the machines, machine state, downtime, OEE, labor and quality aren't recorded yet. A report can't be corrected or reversed yet; a lot it made can be written off on its page.
- **Batches:** the batch size is per material, not per line (a 50 kg mixer next to a 200 kg one isn't modelled). A replacement job is always rounded up to whole batches; to make fewer, cancel it. Orders from the ERP aren't imported yet, so there is no rule yet for an ERP order that isn't whole batches. A rejected partial batch shows in the job's reports as a correction (good down, scrap up).
- **No screen for the reporting tolerances:** the rate and consumption tolerances (110% by default) are set per site through the API.
- **Inventory is declared by hand.** Consumption and production are entered by people; loading lots at feed points, consumption by time window, backflush and scanning aren't built. Genealogy is per job (see Inventory, in detail).
- **Not in inventory yet:** pallets or containers that hold several lots, lot labels and barcode scanning, lot attributes and certificates of analysis, setting stock aside for a job, and suggestions of which lot to use first. Movements can't be back-dated.
- **Reversing is limited** to a lot's latest movement, and not consumption, production, splits, merges, transfers between balances or corrections: put those right with another movement.
- **Reason trees** (a reason grouped under another) can't be edited.
- **Where lots mix, genealogy is approximate:** every lot that ever joined a silo's balance counts as a parent of everything taken from it afterwards. Residence time and first-in, first-out through a silo come with machine data.
- **Inventory movements aren't sent to the ERP yet:** they are recorded as events, but the connection that delivers them isn't built.
- **Shortages use published times, not the live schedule,** and don't count incoming deliveries.
- **Process links don't drive the schedule yet:** a Pack job can be projected to start before the Extrude job it follows. On the floor the link is enforced: the Pack job can't start until the link is met.
- **The Overview** shows no machine state, counts, OEE, downtime, quality, lots or labor yet: those parts are hatched placeholders until the data exists. Its line strip is a rolling 24 hours rather than the current shift.
- **Notes** can only be written when the order is created.
- **Signing in through your company's identity provider** (Okta, Entra ID, AD FS, Keycloak) isn't built yet: users have Pulaski passwords, and an administrator resets forgotten ones (Pulaski sends no e-mail).
- **Reported production and consumption can't be corrected** by anyone yet, so the Auditor role can't do that part of its job; write a lot off or up instead.

- **The schedule shows its buttons to everyone who can see it;** someone who can see the schedule but not plan sees them, and a click explains that planning needs permission instead of changing anything. Switched-off buttons elsewhere can look like ordinary ones.
- **Only Administrator is held everywhere;** every role you create is held at a site.
- **The site picker** on a screen lists every site where you can see something, not only the sites where you can see that screen; choosing one where you can't gives a "may not see this at this site" message.

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22. Troubleshooting and questions

"You may not do this…" or a button I expected isn't there. Your roles don't allow that change, or not at this site. The message names the permission; ask an administrator (see What each role can change).

I can't sign in. Check the user name and password (the password is case-sensitive). After five wrong tries you are locked out for 15 minutes; an administrator can reset your password, which unlocks you. A user who is switched off or has no role can't sign in.

"Your session ended". Sessions last up to 12 hours, and end straight away if an administrator switches you off. Sign in again.

I can't find a process when creating an order. The process must be at the order's site, be active, have a released version, and make the material (all materials, its class, or the material itself). If it is listed but greyed out, read the reason under the field: usually a missing capability.

The schedule is empty. Mark lines as **Schedulable** (and **Can run jobs**) on the Equipment page. Then create a work order.

My new job is at the end of the line, but I want it sooner. That is where new jobs go. Drag it earlier on the Gantt, or use **Line and sequence** in its side panel.

I dragged a job to an earlier day on the Calendar and nothing changed. Dragging earlier only clears a hold. The sequence on the line decides when it runs: move it up the line on the Gantt.

A job won't move. Check its side panel: it may be **locked**, **dispatched** (un-dispatch the order first), **firmed** (it can change line but keeps its time) or already running.

The job's bar looks the wrong length. Its length comes from the rate (run standard) for that material on that line; *No run standard* means there is none. Check the **Rate** in its side panel.

Times look a few hours off. Times are in the site's time zone, not your computer's.

I see "changed by someone else". Another person saved first. Reload, check what they did, and make your change again.

I can't cancel a work order. Cancel its open jobs first (from each job's side panel), then the order.

"…doesn't convert to…" when creating an order or saving a BOM. Pulaski doesn't know how this material converts between those units. Open **Units**, choose the material, and add a conversion (for example 1 m = 1.6 kg). Then try again.

I can't receive a lot: "No location holds inventory". Tick **Holds inventory** on a store or staging area on the Equipment page.

"Lot … has only 12 kg". You tried to take more than the lot holds. Check the quantity and unit, or write the lot up first if the count was wrong.

I can't consume a lot. Only an **available** lot with stock can be consumed. Release a held lot first; a used-up lot has nothing left.

"Choose an adjustment reason" with an empty list. No reasons are set up: ask your administrator to load `db/seed-reasons.sql`.

"Only … latest movement can be reversed". Someone recorded another movement on the lot since you opened it. Reload the lot: **Reverse** shows on the row that can be undone.

I can't post a count. Every lot must have a count (enter 0 if it isn't there), the counts must be saved, and a lot marked *changed since counted* must be counted again.

I can't merge two lots. They must be the same material, at the same location, and in the same status. Move or release one first.

A BOM won't release. Read the yellow box: a component that doesn't exist or is inactive, an operation number the process's released version doesn't have, a unit with no conversion, or no components at all. Save the draft after fixing.

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23. Glossary

Term	Meaning
Balance	The one quantity of a material at a location, with no lot number: for a material that isn't lot-tracked, or at a location where lots mix.
Base unit	The unit a material is stocked in; also the base unit of a kind of unit (kg, m, ea…).
Baseline	A job's times as published when its order was dispatched; the plan you're measured against.
Batch	A fixed amount of a material made at a time, such as a 50 kg mix. Orders of a batch material are whole batches.
BOM	Bill of materials: what one material is made from, per a base quantity.
Capability	A line's permission to run a material, with its ideal rate.
Changeover	Time to switch a line from one product to another.
Crew	A team of people who work the same shifts. A shift schedule names the crew on each shift.
Conversion	How a quantity in one unit becomes another: standard for units of one kind, or set per material.
Count	Checking what is really at a location, then posting the differences.
Department / group	Equipment you can plan to that doesn't run jobs itself; its lines do.
Dispatch	Publish an order to the floor, fixing its baseline.
Draft / released / obsolete	Versions of a process or BOM: being edited / in use for new orders / replaced.
Edge node	Pulaski Edge, running near the machines: reads them, buffers what it reads and forwards it to Pulaski.
Firm	Freeze an order's plan; its jobs keep their start times.
Genealogy	Which lots were made from which.
Hold / quarantine	Statuses that keep a lot on hand but stop it being consumed until released.
Hold until	The earliest a job may start.
Job	The part of a work order on one line (or department).
Journal	The list of every lot movement in a site.
Line	Equipment that runs jobs, such as an extrusion line.
Location	Equipment that holds inventory: a store, staging area or feed point.
Lock	Pin a job so it keeps its place.
Lot	One numbered batch of a material, with its quantity, location and history.
Machine profile	How one kind of machine reports itself: what its state values mean in PackML, and its own stop, alarm and warning codes with their utilization reasons.
Material class	A group of similar materials, such as *Extruded profiles*.
Material type	A label from the item master, such as its MRP class (raw, finished, packaging…); it drives no rules.
Origin	Where a material is maintained: in Pulaski, or in the ERP, which then owns its code, name, base unit and type.
Merge	Combine lots of one material into one.
Micro-stop	A stop shorter than the equipment's threshold (5 minutes by default): logged, but not coded and not counted against availability.
Operation	One step of a process, run on one or more lines or departments.

Term	Meaning
Packaging unit	A unit like box, bag or pallet whose size depends on the material; it converts only through a material conversion.
Partial batch	A batch cut short when the line stopped. Accepted for work in progress; held for approval for finished goods.
Process (route)	The operations an order goes through, and where each runs.
Projection	Pulaski's calculation of when each job will run, redone on every read.
Rate check	A run can't hold more than its length × the line's ideal rate × the site's tolerance.
Ready	A job that can start: its order is on the floor and its process links are met.
Reverse	Undo a lot's latest movement.
Permission	One kind of change Pulaski lets people make, such as *Receive lots*. Roles hold permissions.
Role	A set of permissions, such as Planner or Quality, that an administrator maintains and gives to users at a site (Administrator everywhere). See Users, roles and signing in.
Run	One stretch of a job running on a line, from a start to a pause or completion. Production and consumption go into a run.
Labor session	A stretch of time one person spent on a line: direct, setup, indirect or training. See Labor.
Shift schedule	When a site works: shifts repeating over a cycle of days. Equipment follows one, inherits its parent's or runs around the clock. See Shift schedules.
Run standard (planning rate)	The rate Pulaski plans with for a material on a line: calculated from the ideal rate, or modified by a person.
Scrap factor	Extra component quantity planned for losses.
Sequence	The order of jobs on a line.
Share	The part of an operation's quantity one line or department takes.
Ship	Send stock out of the site.
Shortage	What open jobs still need of a component beyond the stock available.
Site	A plant, with its own time zone.
Split	Divide part of a lot into a new lot.
Stop	Time a machine isn't executing, from leaving Execute until it executes again, with the reason it is coded with.
Substitution	Consuming a material a job had no planned component of.
Trace	Following genealogy back to everything a lot was made from, or forward to everything made from it.
Sparkplug B	A standard way for machines to publish their data over MQTT, with birth certificates naming each metric.
Tag	A machine signal bound to equipment, with the role it plays (state, a count…).
Utilization reason	What a machine's time counts as: *runtime*, *downtime* (against availability) or *planned* (left out of OEE).
Where used	The BOMs and open jobs that use a material as a component.
Window	The time span a department job must be run in.
Work order	A request to make a quantity of one material by a due date.
Write up / write off	Add found stock to a lot / remove damaged, lost or miscounted stock, each for a reason.